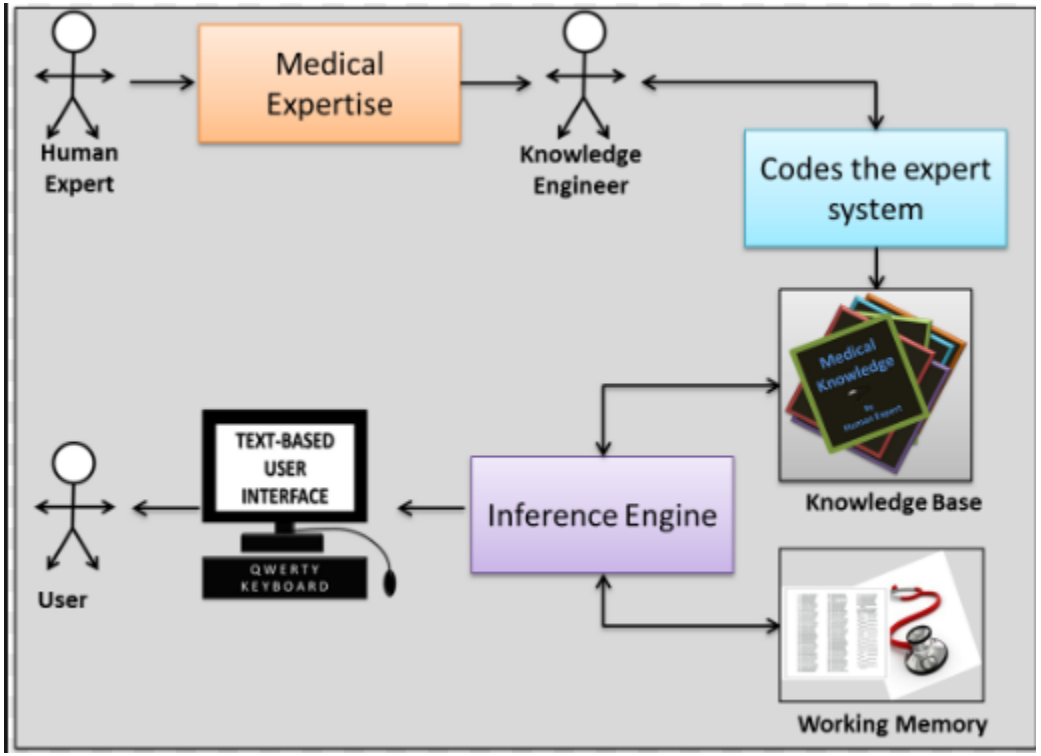
Uncertainty and Fuzzy Expert System



Objectives

- Understand what is uncertainty and how fuzzy logic can be used to solve uncertainty;
- Construct a simple system using fuzzy logic toolbox in Matlab.

UNCERTAINTY



What is uncertainty?

- It is defined as the lack of the exact knowledge that would enable us to reach a perfectly reliable conclusion.

Sources of uncertain knowledge

1) Weak implications.
2) Imprecise language.
3) Unknown data.
4) Combining the views of different experts.

Weak implications. Domain experts and knowledge engineers have the painful task of establishing concrete correlations between IF (condition) and THEN (action) parts of the rules. Therefore, expert systems need to have the ability to handle vague associations, for example by accepting the degree of correlations as numerical certainty factors.

Imprecise language. Our natural language is ambiguous and imprecise. We describe facts with such terms as *often* and *sometimes*, *frequently* and *hardly ever*. As a result, it can be difficult to express knowledge in the precise IF-THEN form of production rules. However, if the meaning of the facts is quantified, it can be used in expert systems. In 1944, Ray Simpson asked 355 high school and college students to place 20 terms like *often* on a scale between 1 and 100. In 1968, Milton Hakel repeated this experiment.

Unknown data. When the data is incomplete or missing, the only solution is to accept the value "unknown" and proceed to an approximate reasoning with this value.

Combining the views of different experts. Large expert systems usually combine the knowledge and expertise of a number of experts. Unfortunately, experts often have contradictory opinions and produce conflicting rules. To resolve the conflict, the knowledge engineer has to attach a weight to each expert and then calculate the composite conclusion. But no systematic method exists to obtain these weights.

How can an expert system handle uncertainty?

- Experts rely on common sense when they solve problems which may be uncertain where they may use vague and ambiguous terms.
- Expert system can handle uncertainty using fuzzy logic and we can call it fuzzy expert system.

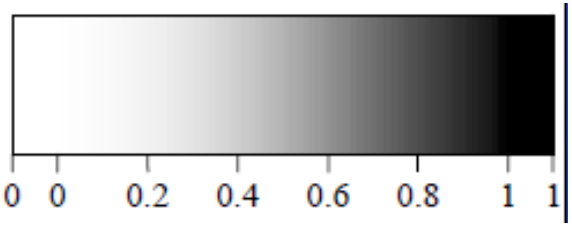
What is fuzzy logic?

- Fuzzy logic describes fuzziness that involves the theory of fuzzy sets.
- It reflects how people think and attempts to model our sense of words, decision making and common sense.

What is the difference between boolean logic and fuzzy logic?

- Boolean or crisp logic is a two-valued logic that uses sharp distinction where a line can be drawn between members of a class and non-members.
- Fuzzy logic is a multi-values logic that employs the spectrum of colours, accepting that things can be partly true and false at the same time.



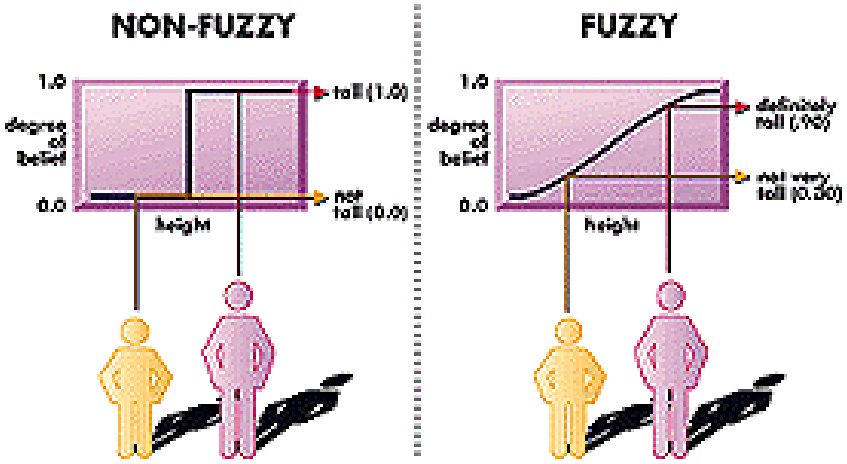


Fuzzy logic…

- Fuzzy logic is a set of mathematical principles for knowledge representation based on degrees of membership.

Fuzzy logic…

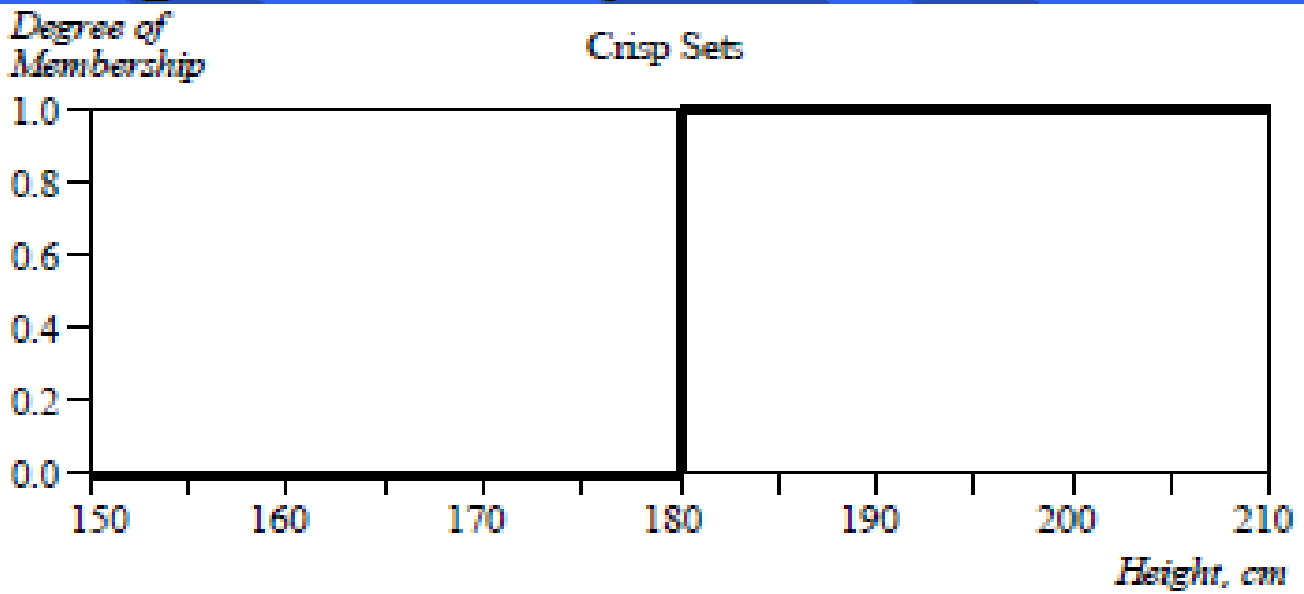
- The classical example in fuzzy sets is ‘Tall men’.
- The elements of the fuzzy set ‘Tall men’ are all men but their degree of membership depend on their height.

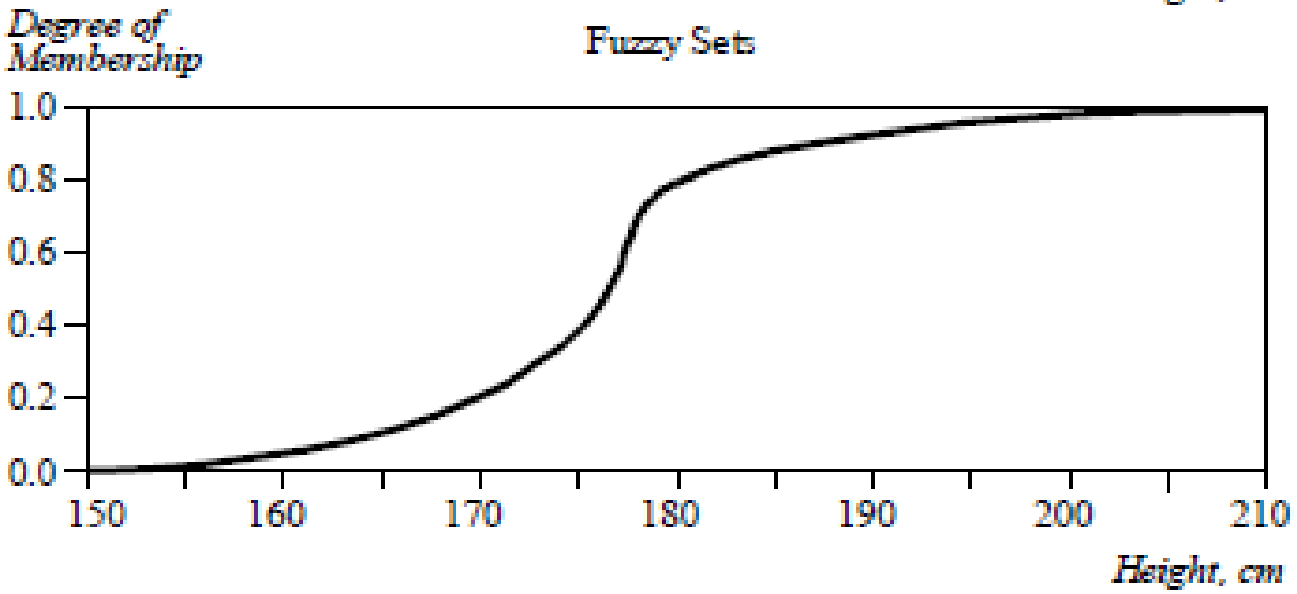


Sample fuzzy set for 'Tall men'

Name	Height, cm	Degree of Membership	
		Crisp	*Fuzzy*
Chris	208	1	1.00
Mark	205	1	1.00
John	198	1	0.98
Tom	181	1	0.82
David	179	0	0.78
Mike	172	0	0.24
Bob	167	0	0.15
Steven	158	0	0.06
Bill	155	0	0.01
Peter	152	0	0.00

Crisp and fuzzy sets of "*tall men*"





Fuzzy logic (refer to previous figure)

- The *x*-axis represents the **universe of discourse** – the range of all possible values applicable to a chosen variable. In our case, the variable is the man height. According to this representation, the universe of men's heights consists of all tall men.
- The *y*-axis represents the **membership value of the fuzzy set**. In our case, the fuzzy set of "*tall men*" maps height values into corresponding membership values.

A fuzzy set is a set with fuzzy boundaries.

- Let X be the universe of discourse and its elements be denoted as x. In the classical set theory, **crisp set A of X is defined as function $f_A(x)$ called the characteristic function of A**

$f_A(x): X \rightarrow \{0, 1\}$, where $f_A(x) = \begin{cases} 1, \text{if } x \in A \\ 0, \text{if } x \notin A \end{cases}$

This set maps universe X to a set of two elements. For any element x of universe X, characteristic function $f_A(x)$ is equal to 1 if x is an element of set A, and is equal to 0 if x is not an element of A.

- In the fuzzy theory, fuzzy set A of universe X is defined by function $\mu_A(x)$ called the *membership function* of set A

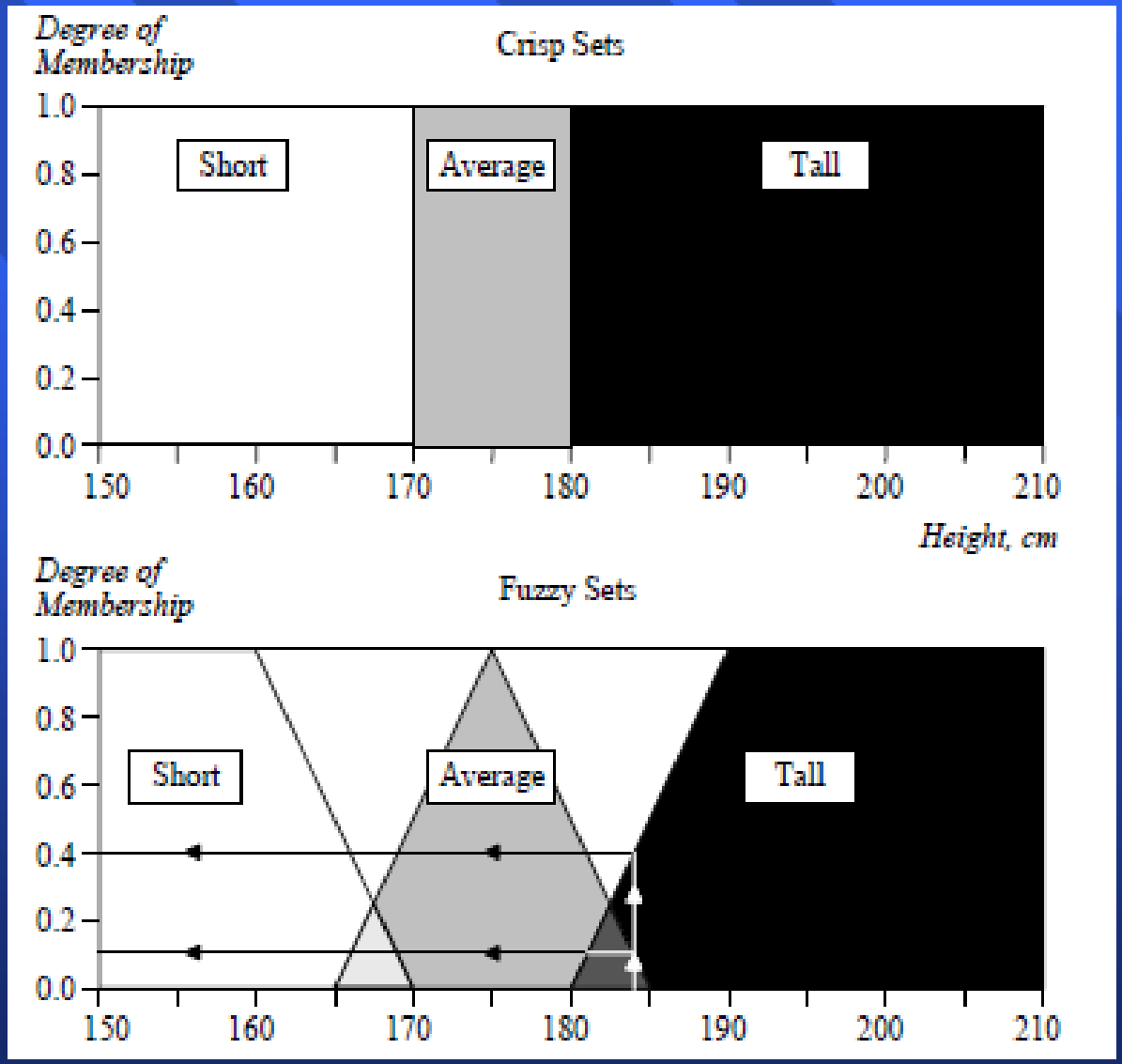
  $\mu_A(x)$: $X \rightarrow [0, 1]$, where $\mu_A(x) = 1$ if x is totally in A;
  $\mu_A(x) = 0$ if x is not in A;
  $0 < \mu_A(x) < 1$ if x is partly in A.

  This set allows a continuum of possible choices. For any element x of universe X, membership function $\mu_A(x)$ equals the degree to which x is an element of set A. This degree, a value between 0 and 1, represents the **degree of membership**, also called **membership value**, of element x in set A.

How to represent a fuzzy set in a computer?

- First, we determine the membership functions. In our "*tall men*" example, we can obtain fuzzy sets of *tall*, *short* and *average* men.
- The universe of discourse – the men's heights – consists of three sets: *short*, *average* and *tall men*. As you will see, a man who is 184 cm tall is a member of the *average men* set with a degree of membership of 0.1, and at the same time, he is also a member of the *tall men* set with a degree of 0.4.

Crisp and fuzzy sets of short, average and tall men



Linguistic variables and hedges

- At the root of fuzzy set theory lies the idea of linguistic variables.
- **A linguistic variable is a fuzzy variable.** For example, the statement "John is tall" implies that the linguistic variable *John* takes the linguistic value *tall*.

In fuzzy expert systems, linguistic variables are used in fuzzy rules. For example:

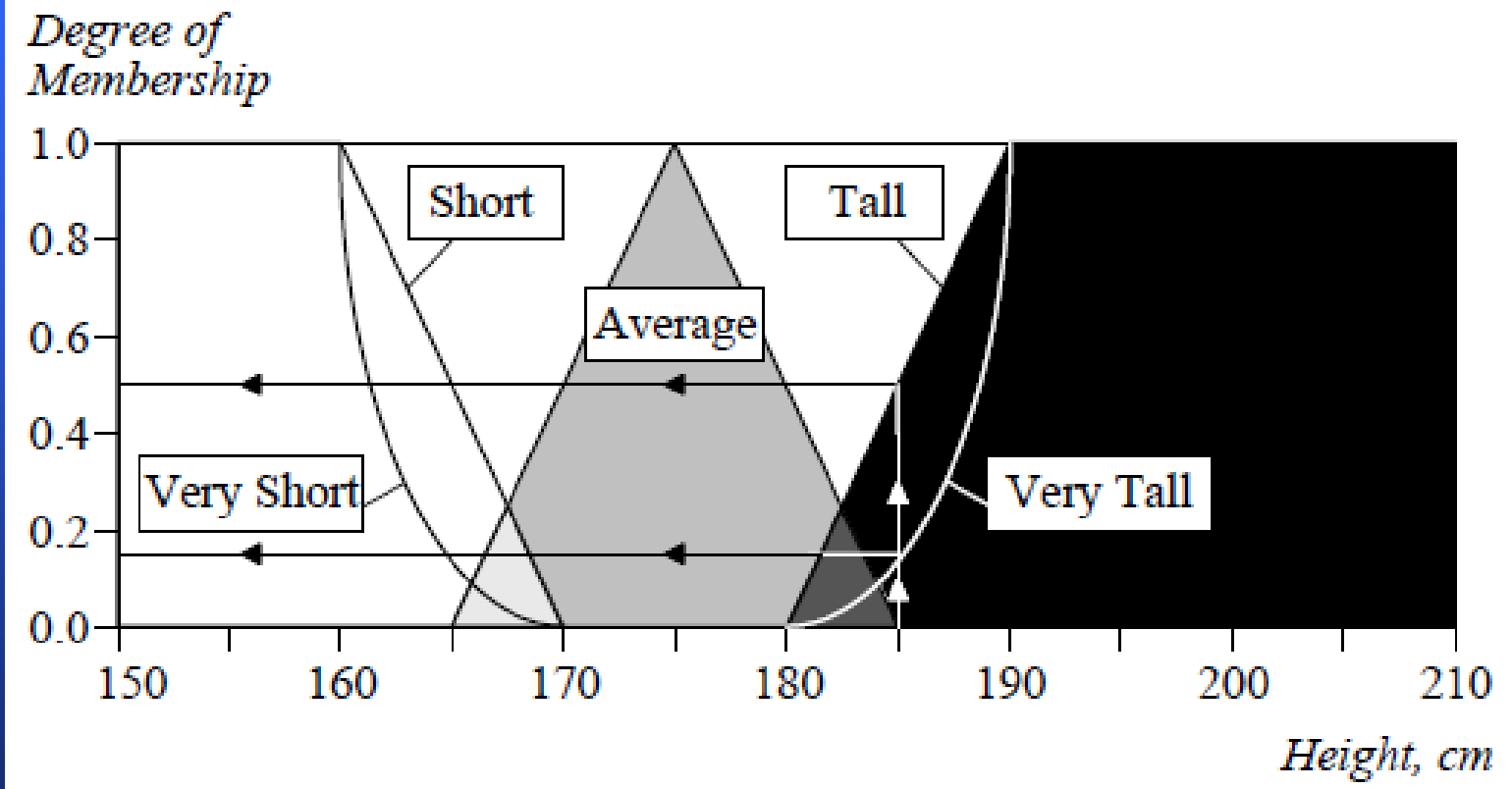
IF wind is strong
THEN sailing is good

IF project_duration is long
THEN completion_risk is high

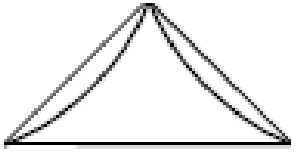
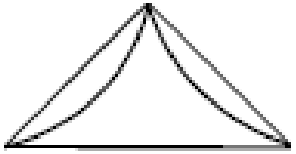
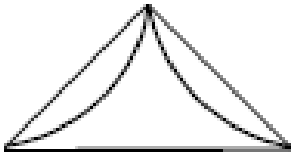
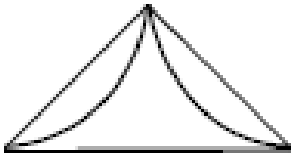
IF speed is slow
THEN stopping_distance is short

- The range of possible values of a linguistic variable represents the universe of discourse of that variable. For example, the universe of discourse of the linguistic variable *speed* might have the range between 0 and 220 km/h and may include such fuzzy subsets as *very slow*, *slow*, *medium*, *fast*, and *very fast*.
- A linguistic variable carries with it the concept of fuzzy set qualifiers, called *hedges*.
- **Hedges are terms that modify the shape of fuzzy sets. They include adverbs such as *very*, *somewhat*, *quite*, *more or less* and *slightly*.**

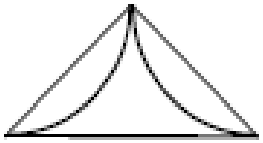
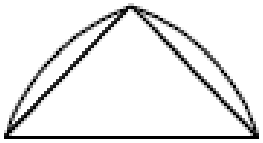
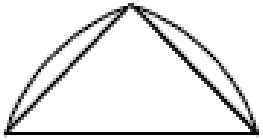
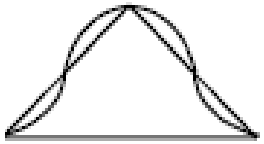
Fuzzy sets with the hedge *very*



Representation of hedges in fuzzy logic

Hedge	*Mathematical Expression*	*Graphical Representation*
A little	$[\mu_A(x)]^{1.3}$	
Slightly	$[\mu_A(x)]^{1.7}$	
Very	$[\mu_A(x)]^{2}$	
Extremely	$[\mu_A(x)]^{3}$	

Representation of hedges in fuzzy logic (continued)

Hedge	*Mathematical Expression*	*Graphical Representation*
Very very	$[\mu_A(x)]^4$	
More or less	$\sqrt{\mu_A(x)}$	
Somewhat	$\sqrt{\mu_A(x)}$	
Indeed	$2\,[\mu_A(x)]^2$ if $0 \le \mu_A \le 0.5$ $1 - 2\,[1 - \mu_A(x)]^2$ if $0.5 < \mu_A \le 1$	

What is a fuzzy rule?

A fuzzy rule can be defined as a conditional statement in the form:

IF *x* **is** *A*
THEN *y* **is** *B*

where *x* and *y* are linguistic variables; and *A* and *B* are linguistic values determined by fuzzy sets on the universe of discourses *X* and *Y*, respectively.

What is the difference between classical and fuzzy rules?

A classical IF-THEN rule uses binary logic, for example,

Rule: 1
IF speed is > 100
THEN stopping_distance is long

Rule: 2
IF speed is < 40
THEN stopping_distance is short

The variable *speed* can have any numerical value between 0 and 220 km/h, but the linguistic variable *stopping_distance* can take either value *long* or *short*. In other words, classical rules are expressed in the black-and-white language of Boolean logic.

We can also represent the stopping distance rules in a fuzzy form:

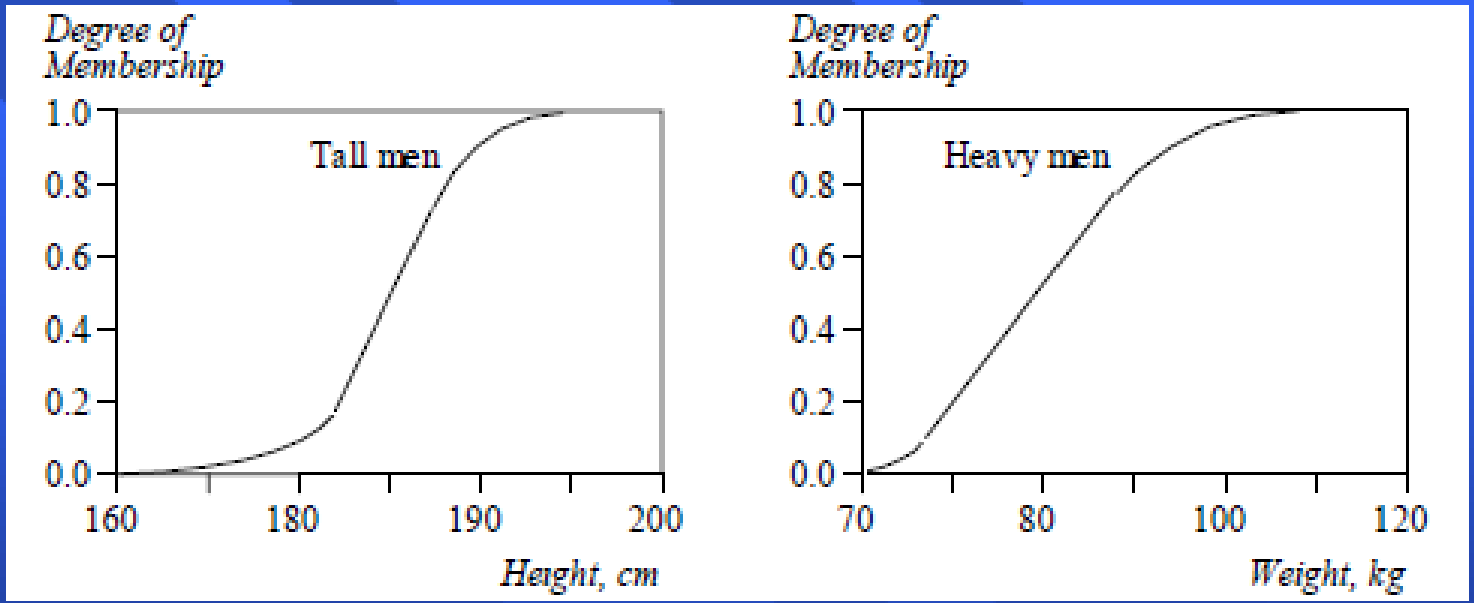
Rule: 1
IF speed is fast
THEN stopping_distance is long

Rule: 2
IF speed is slow
THEN stopping_distance is short

In fuzzy rules, the linguistic variable *speed* also has the range (the universe of discourse) between 0 and 220 km/h, but this range includes fuzzy sets, such as *slow*, *medium* and *fast*. The universe of discourse of the linguistic variable *stopping_distance* can be between 0 and 300 m and may include such fuzzy sets as *short*, *medium* and *long*.

- Fuzzy rules relate fuzzy sets.
- In a fuzzy system, all rules fire to some extent, or in other words they fire partially. If the antecedent is true to some degree of membership, then the consequent is also true to that same degree.

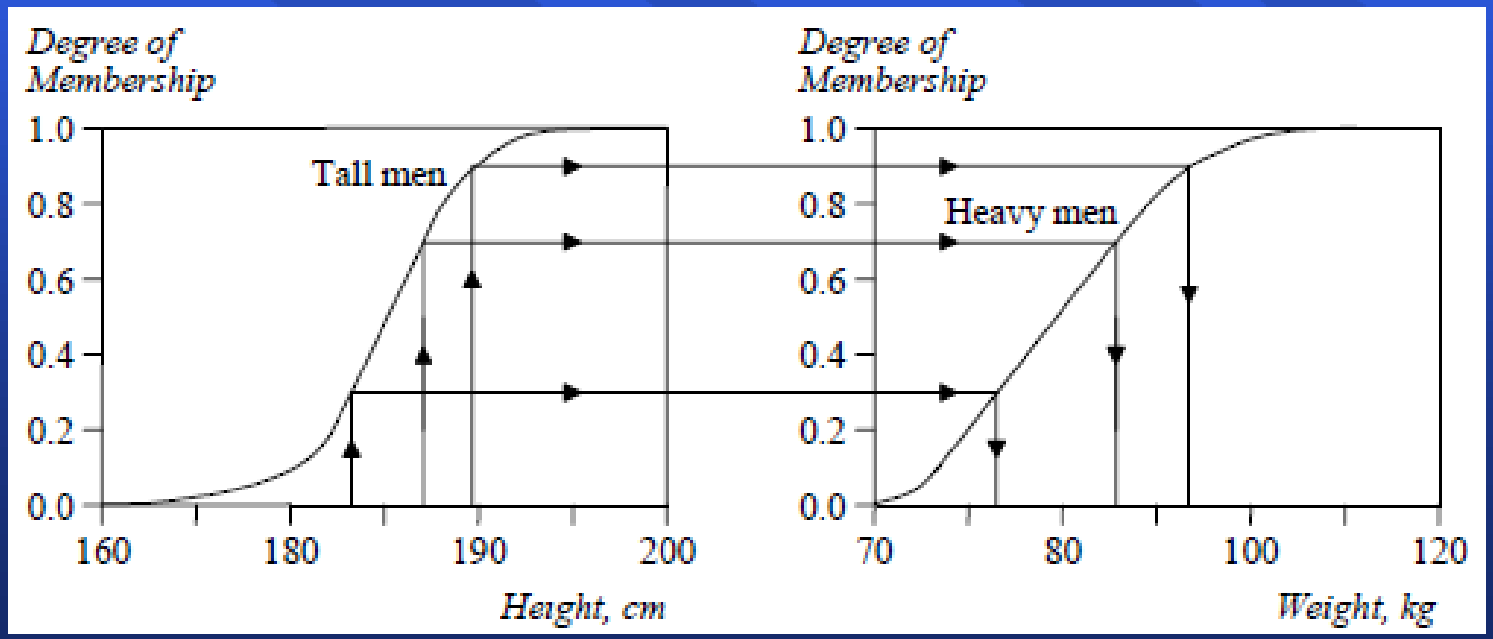
Fuzzy sets of *tall* and *heavy* men



These fuzzy sets provide the basis for a weight estimation model. The model is based on a relationship between a man's height and his weight:

IF **height is *tall***
THEN **weight is *heavy***

The value of the output or a truth membership grade of the rule consequent can be estimated directly from a corresponding truth membership grade in the antecedent. This form of fuzzy inference uses a method called **monotonic selection**.



A fuzzy rule can have multiple antecedents, for example:

IF project_duration is long
AND project_staffing is large
AND project_funding is inadequate
THEN risk is high

IF service is excellent
OR food is delicious
THEN tip is generous

The consequent of a fuzzy rule can also include multiple parts, for instance:

IF temperature is hot
THEN hot_water is reduced;
cold_water is increased

Fuzzy inference

The most commonly used fuzzy inference technique is the so-called Mamdani method. In 1975, Professor **Ebrahim Mamdani** of London University built one of the first fuzzy systems to control a steam engine and boiler combination. He applied a set of fuzzy rules supplied by experienced human operators.

Mamdani fuzzy inference

- The Mamdani-style fuzzy inference process is performed in four steps:
  - fuzzification of the input variables,
  - rule evaluation;
  - aggregation of the rule outputs, and finally
  - defuzzification.

We examine a simple two-input one-output problem that includes three rules:

Rule: 1
IF *x* **is** *A*3
OR *y* **is** *B*1
THEN *z* **is** *C*1

Rule: 2
IF *x* **is** *A*2
AND *y* **is** *B*2
THEN *z* **is** *C*2

Rule: 3
IF *x* **is** *A*1
THEN *z* **is** *C*3

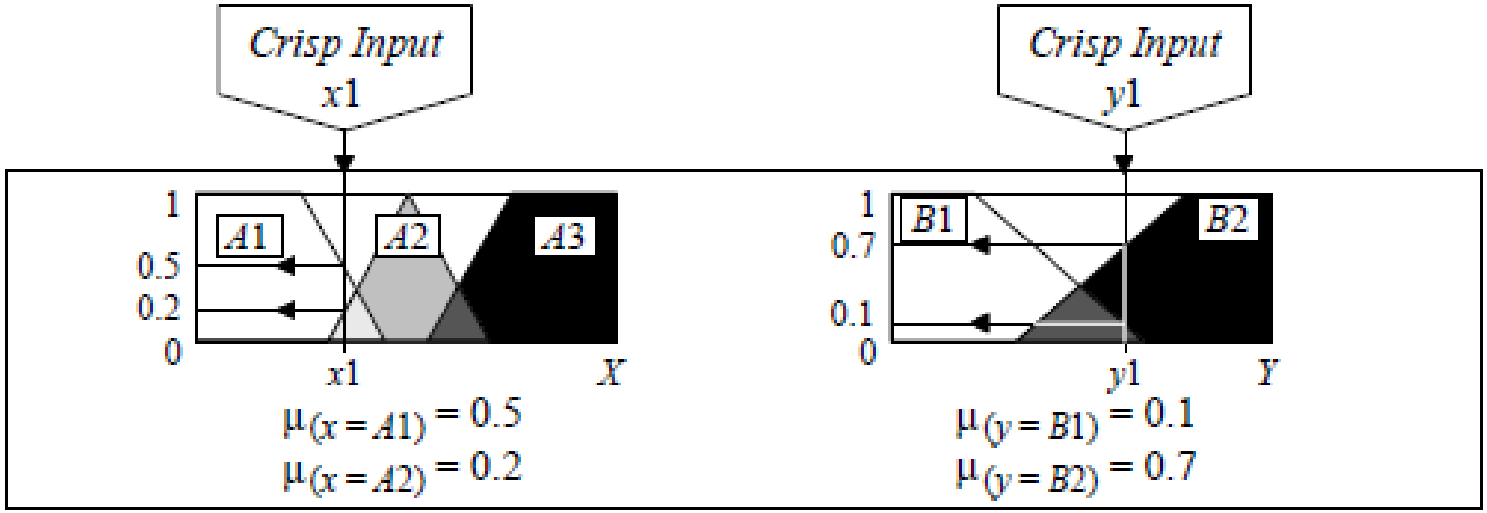
Rule: 1
IF *project_funding* **is** *adequate*
OR *project_staffing* **is** *small*
THEN *risk* **is** *low*

Rule: 2
IF *project_funding* **is** *marginal*
AND *project_staffing* **is** *large*
THEN *risk* **is** *normal*

Rule: 3
IF *project_funding* **is** *inadequate*
THEN *risk* **is** *high*

Step 1: Fuzzification

The first step is to take the crisp inputs, $x1$ and $y1$ (*project funding* and *project staffing*), and determine the degree to which these inputs belong to each of the appropriate fuzzy sets.



Step 2: Rule Evaluation

The second step is to take the fuzzified inputs, $\mu_{(x=A1)} = 0.5$, $\mu_{(x=A2)} = 0.2$, $\mu_{(y=B1)} = 0.1$ and $\mu_{(y=B2)} = 0.7$, and apply them to the antecedents of the fuzzy rules. If a given fuzzy rule has multiple antecedents, the fuzzy operator (AND or OR) is used to obtain a single number that represents the result of the antecedent evaluation. This number (the truth value) is then applied to the consequent membership function.

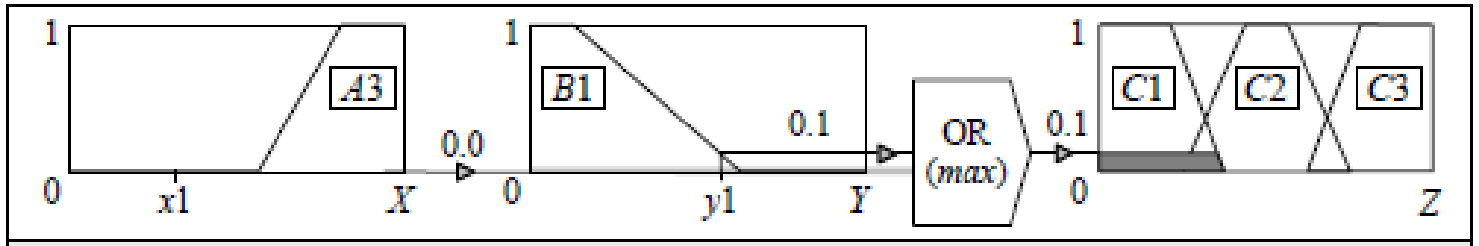
To evaluate the disjunction of the rule antecedents, we use the **OR fuzzy operation**. Typically, fuzzy expert systems make use of the classical fuzzy operation **union**:

$$\mu_{A \cup B}(x) = max\ [\mu_A(x), \mu_B(x)]$$

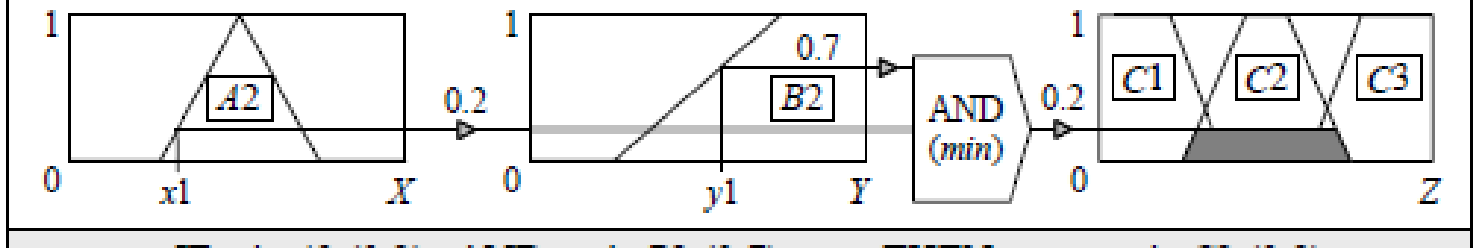
Similarly, in order to evaluate the conjunction of the rule antecedents, we apply the **AND fuzzy operation intersection**:

$$\mu_{A \cap B}(x) = min\ [\mu_A(x), \mu_B(x)]$$

Mamdani-style rule evaluation



Rule 1: IF *x* is *A*3 (0.0) OR *y* is *B*1 (0.1) THEN *z* is *C*1 (0.1)



Rule 2: IF *x* is *A*2 (0.2) AND *y* is *B*2 (0.7) THEN *z* is *C*2 (0.2)



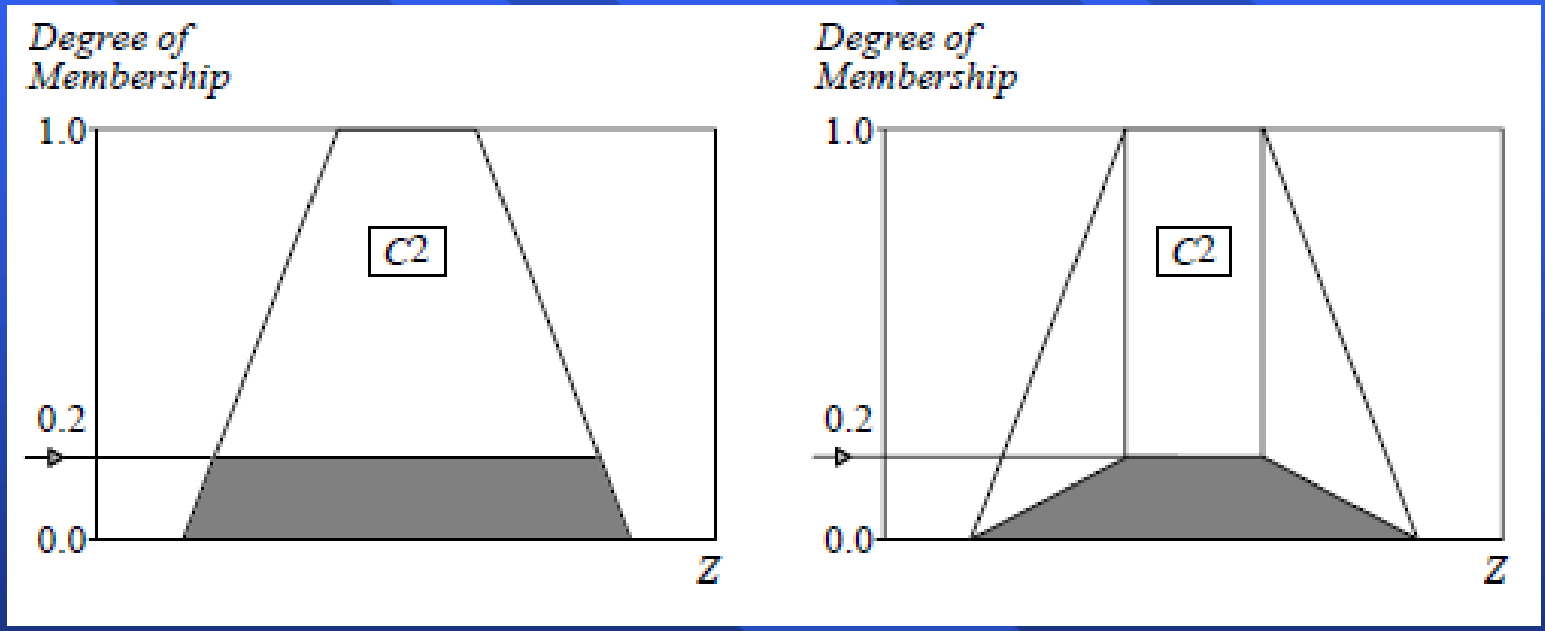
Rule 3: IF *x* is *A*1 (0.5) THEN *z* is *C*3 (0.5)

Now the result of the antecedent evaluation can be applied to the membership function of the consequent.

- The most common method of correlating the rule consequent with the truth value of the rule antecedent is to cut the consequent membership function at the level of the antecedent truth. This method is called **clipping**. Since the top of the membership function is sliced, the clipped fuzzy set loses some information. However, clipping is still often preferred because it involves less complex and faster mathematics, and generates an aggregated output surface that is easier to defuzzify.

- While clipping is a frequently used method, **scaling** offers a better approach for preserving the original shape of the fuzzy set. The original membership function of the rule consequent is adjusted by multiplying all its membership degrees by the truth value of the rule antecedent. This method, which generally loses less information, can be very useful in fuzzy expert systems.

Clipped and scaled membership functions

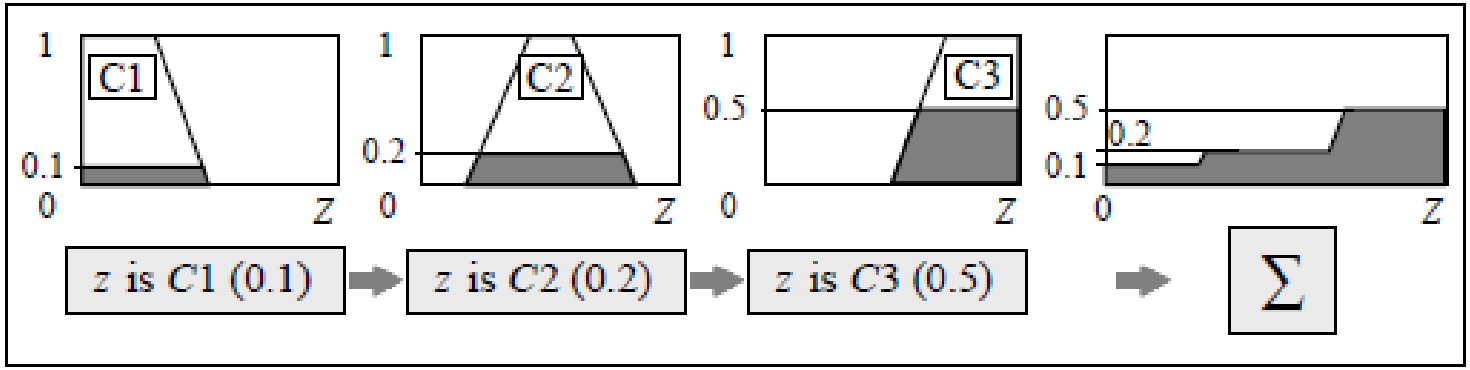


Step 3: Aggregation of the rule outputs

Aggregation is the process of unification of the outputs of all rules. We take the membership functions of all rule consequents previously clipped or scaled and combine them into a single fuzzy set.

The input of the aggregation process is the list of clipped or scaled consequent membership functions, and the output is one fuzzy set for each output variable.

Aggregation of the rule outputs



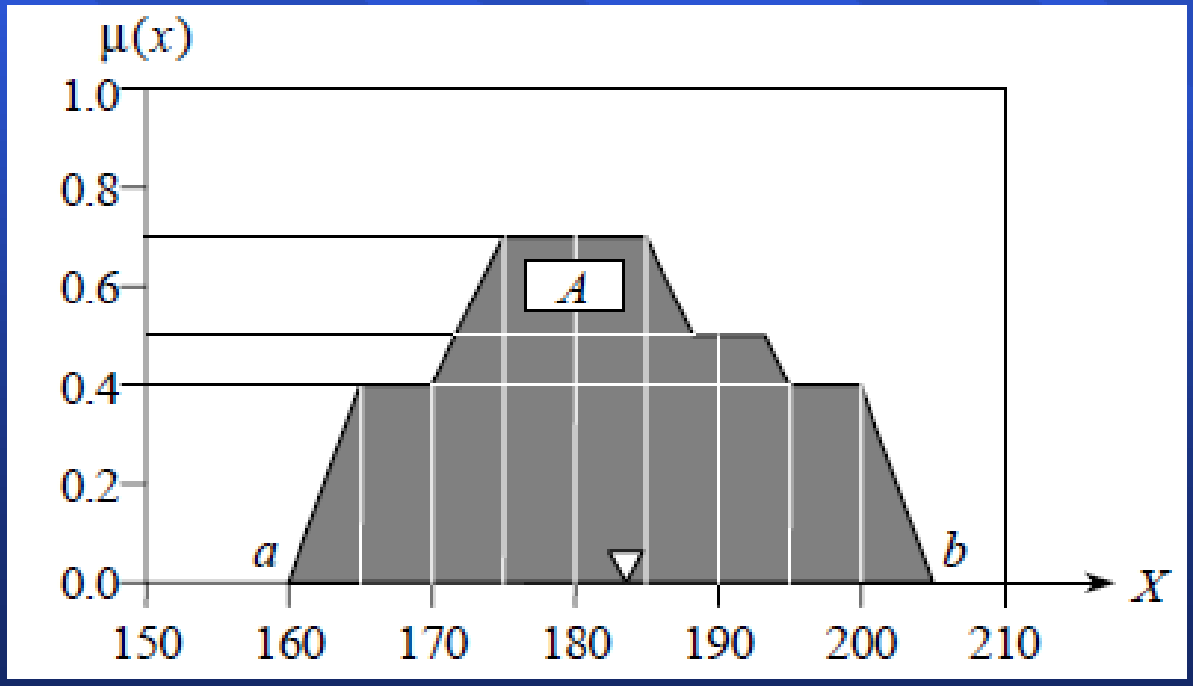
Step 4: Defuzzification

The last step in the fuzzy inference process is defuzzification. Fuzziness helps us to evaluate the rules, but the final output of a fuzzy system has to be a crisp number. The input for the defuzzification process is the aggregate output fuzzy set and the output is a single number.

- There are several defuzzification methods, but probably the most popular one is the **centroid technique**. It finds the point where a vertical line would slice the aggregate set into two equal masses. Mathematically this **centre of gravity (COG)** can be expressed as:

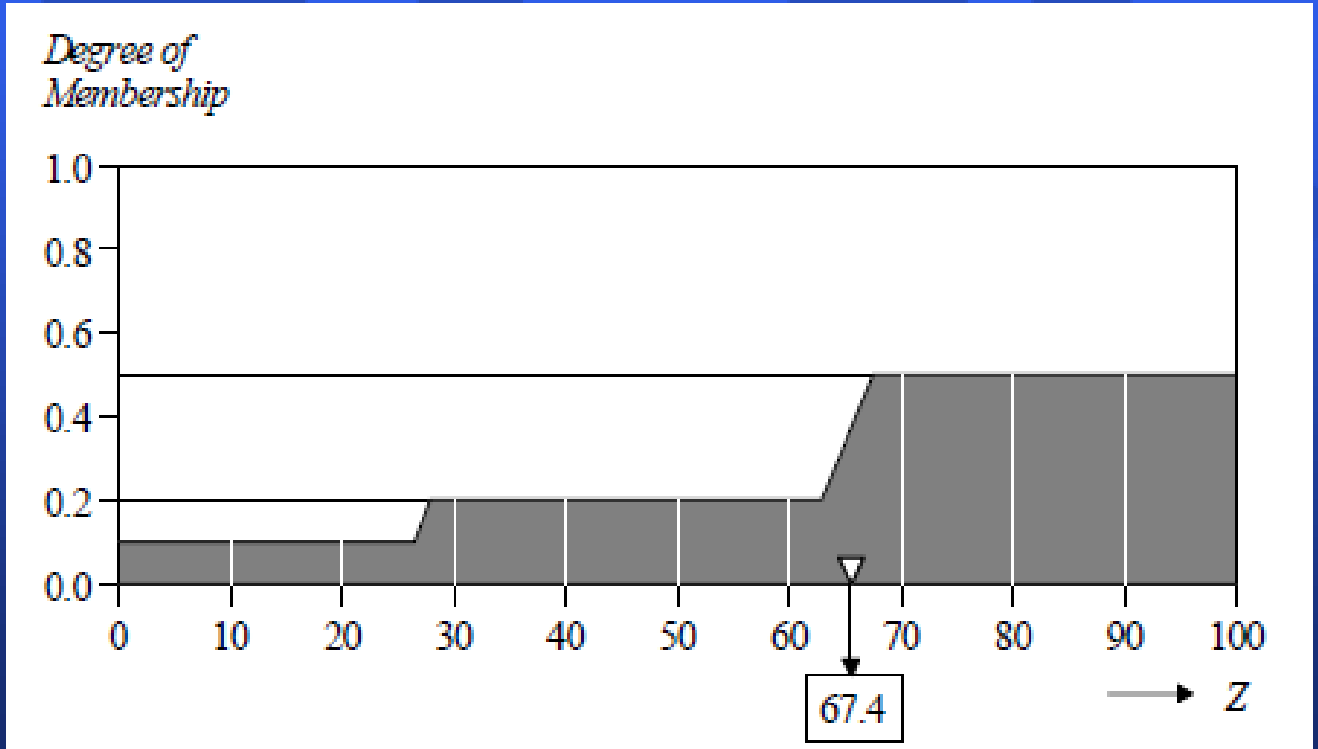
$$COG = \frac{\int_a^b \mu_A(x)\, x\, dx}{\int_a^b \mu_A(x)\, dx}$$

- Centroid defuzzification method finds a point representing the centre of gravity of the fuzzy set, *A*, on the interval, *ab*.
- A reasonable estimate can be obtained by calculating it over a sample of points.



Centre of gravity (COG):

$$COG = \frac{(0+10+20)\times 0.1+(30+40+50+60)\times 0.2+(70+80+90+100)\times 0.5}{0.1+0.1+0.1+0.2+0.2+0.2+0.2+0.5+0.5+0.5+0.5} = 67.4$$



Sugeno fuzzy inference

- Mamdani-style inference, as we have just seen, requires us to find the centroid of a two-dimensional shape by integrating across a continuously varying function. In general, this process is not computationally efficient.
- **Michio Sugeno** suggested to use a single spike, a *singleton*, as the membership function of the rule consequent. A **singleton**, or more precisely a **fuzzy singleton**, is a fuzzy set with a membership function that is unity at a single particular point on the universe of discourse and zero everywhere else.

Sugeno-style fuzzy inference is very similar to the Mamdani method. Sugeno changed only a rule consequent. Instead of a fuzzy set, he used a mathematical function of the input variable. The format of the **Sugeno-style fuzzy rule** is

IF x is A
AND y is B
THEN z is $f(x, y)$

where x, y and z are linguistic variables; A and B are fuzzy sets on universe of discourses X and Y, respectively; and $f(x, y)$ is a mathematical function.

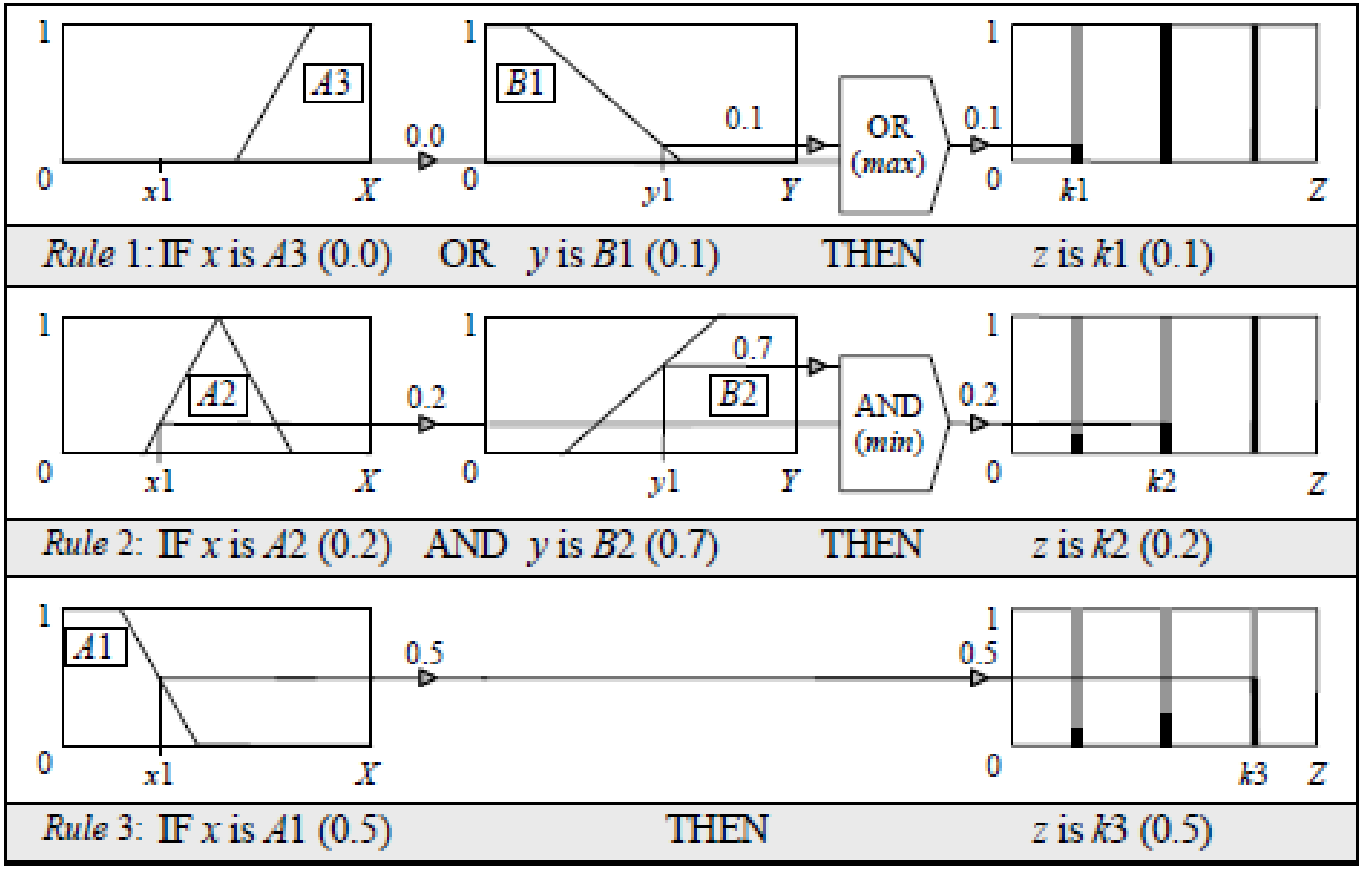
The most commonly used **zero-order Sugeno fuzzy model** applies fuzzy rules in the following form:

IF *x* **is** *A*
AND *y* **is** *B*
THEN *z* **is** *k*

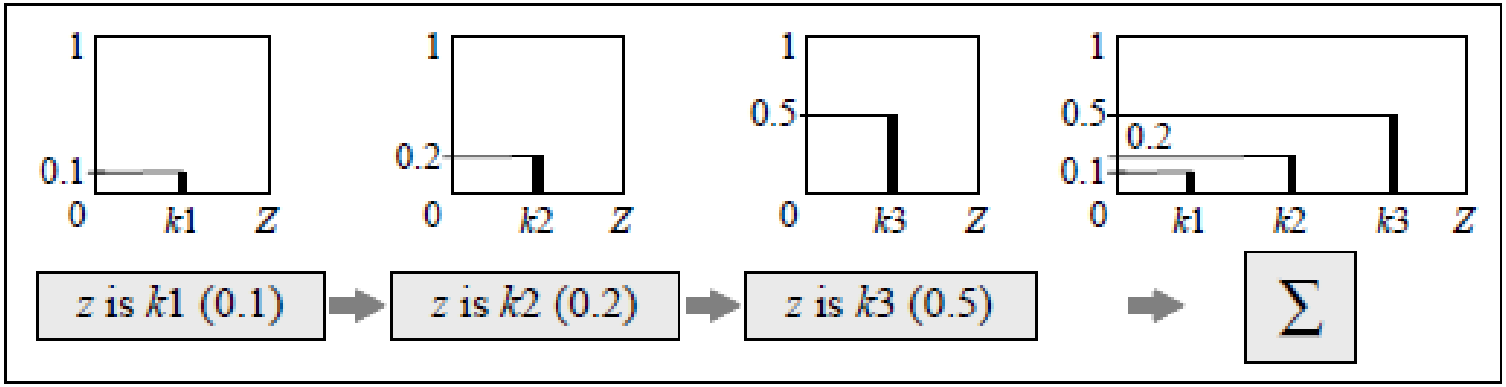
where *k* is a constant.

In this case, the output of each fuzzy rule is constant. All consequent membership functions are represented by singleton spikes.

Sugeno-style rule evaluation



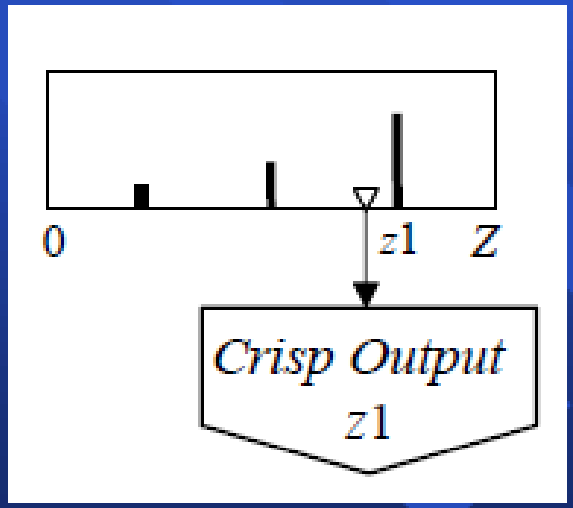
Sugeno-style aggregation of the rule outputs



Weighted average (WA):

$$WA = \frac{\mu(k1)\times k1 + \mu(k2)\times k2 + \mu(k3)\times k3}{\mu(k1) + \mu(k2) + \mu(k3)} = \frac{0.1\times 20 + 0.2\times 50 + 0.5\times 80}{0.1 + 0.2 + 0.5} = 65$$

Sugeno-style defuzzification



How to make a decision on which method to apply – Mamdani or Sugeno?

- Mamdani method is widely accepted for capturing expert knowledge. It allows us to describe the expertise in more intuitive, more human-like manner. However, Mamdani-type fuzzy inference entails a substantial computational burden.
- On the other hand, Sugeno method is computationally effective and works well with optimisation and adaptive techniques, which makes it very attractive in control problems, particularly for dynamic nonlinear systems.

Building a fuzzy expert system: case study

- A service centre keeps spare parts and repairs failed ones.
- A customer brings a failed item and receives a spare of the same type.
- Failed parts are repaired, placed on the shelf, and thus become spares.
- The objective here is to advise a manager of the service centre on certain decision policies to keep the customers satisfied.

Process of developing a fuzzy expert system

1. Specify the problem and define linguistic variables.
2. Determine fuzzy sets.
3. Elicit and construct fuzzy rules.
4. Encode the fuzzy sets, fuzzy rules and procedures to perform fuzzy inference into the expert system.
5. Evaluate and tune the system.

Step 1: Specify the problem and define linguistic variables

There are four main linguistic variables: average waiting time (mean delay) m, repair utilisation factor of the service centre ρ, number of servers s, and initial number of spare parts n.

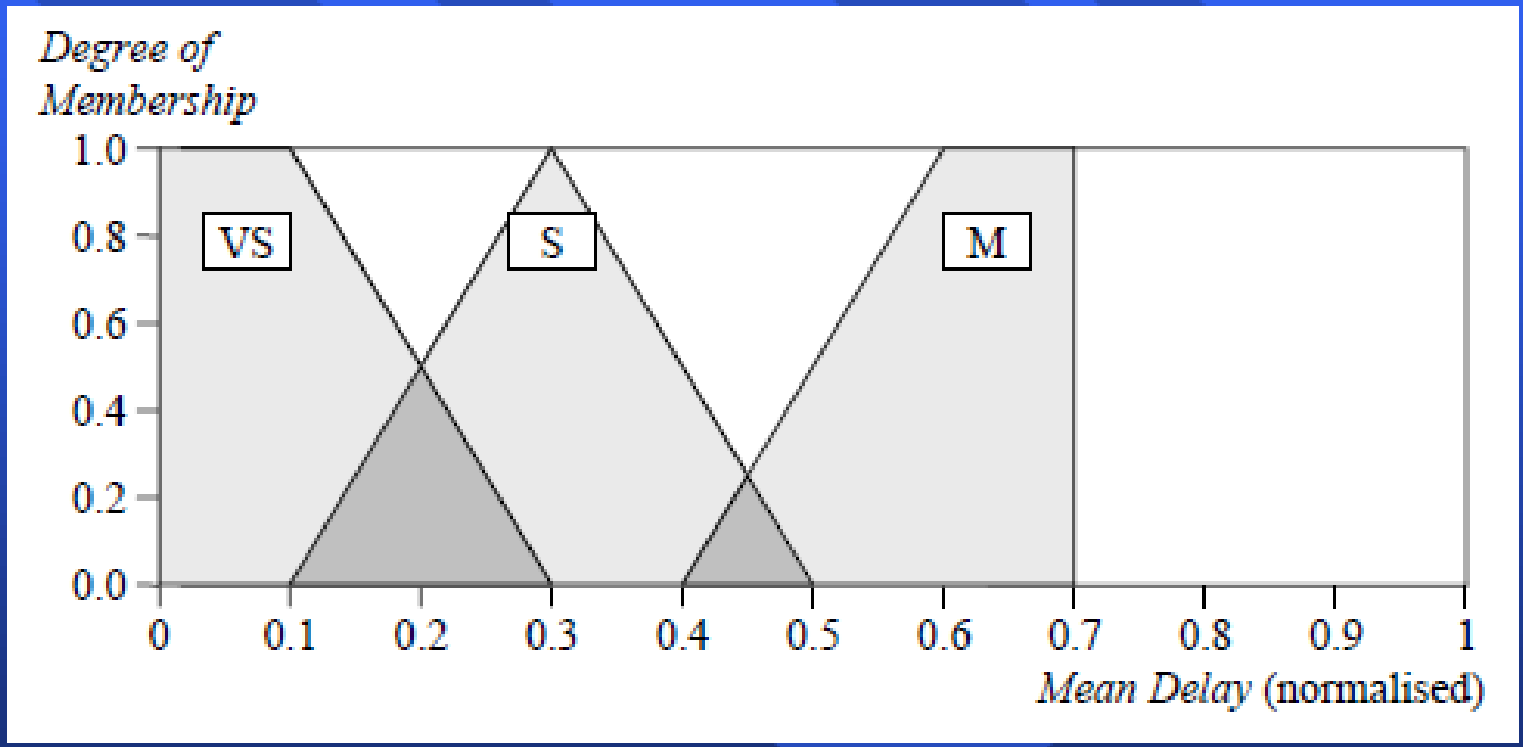
Linguistic variables and their ranges

Linguistic Variable: *Mean Delay*, *m*		
Linguistic Value	Notation	Numerical Range (normalised)
Very Short	VS	[0, 0.3]
Short	S	[0.1, 0.5]
Medium	M	[0.4, 0.7]
Linguistic Variable: *Number of Servers*, *s*		
Linguistic Value	Notation	Numerical Range (normalised)
Small	S	[0, 0.35]
Medium	M	[0.30, 0.70]
Large	L	[0.60, 1]
Linguistic Variable: *Repair Utilisation Factor*, ρ		
Linguistic Value	Notation	Numerical Range
Low	L	[0, 0.6]
Medium	M	[0.4, 0.8]
High	H	[0.6, 1]
Linguistic Variable: *Number of Spares*, *n*		
Linguistic Value	Notation	Numerical Range (normalised)
Very Small	VS	[0, 0.30]
Small	S	[0, 0.40]
Rather Small	RS	[0.25, 0.45]
Medium	M	[0.30, 0.70]
Rather Large	RL	[0.55, 0.75]
Large	L	[0.60, 1]
Very Large	VL	[0.70, 1]

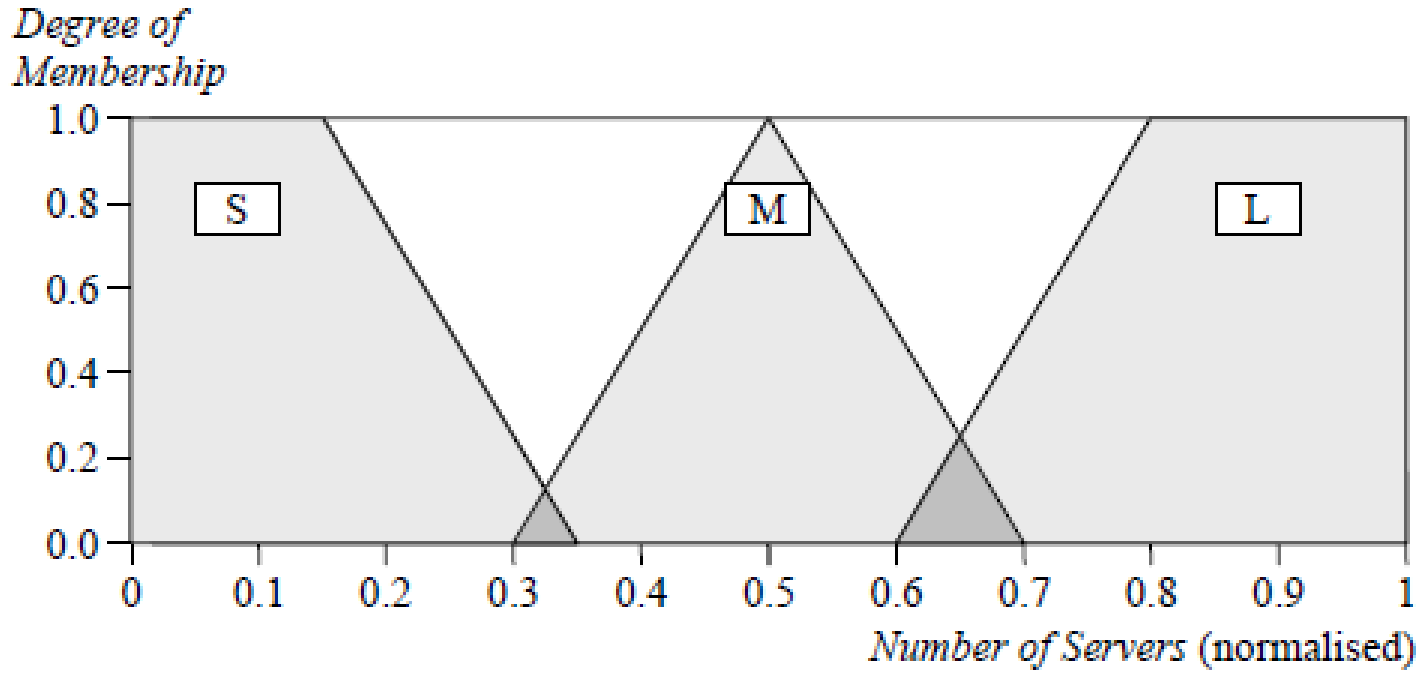
Step 2: Determine fuzzy sets

Fuzzy sets can have a variety of shapes. However, a triangle or a trapezoid can often provide an adequate representation of the expert knowledge, and at the same time, significantly simplifies the process of computation.

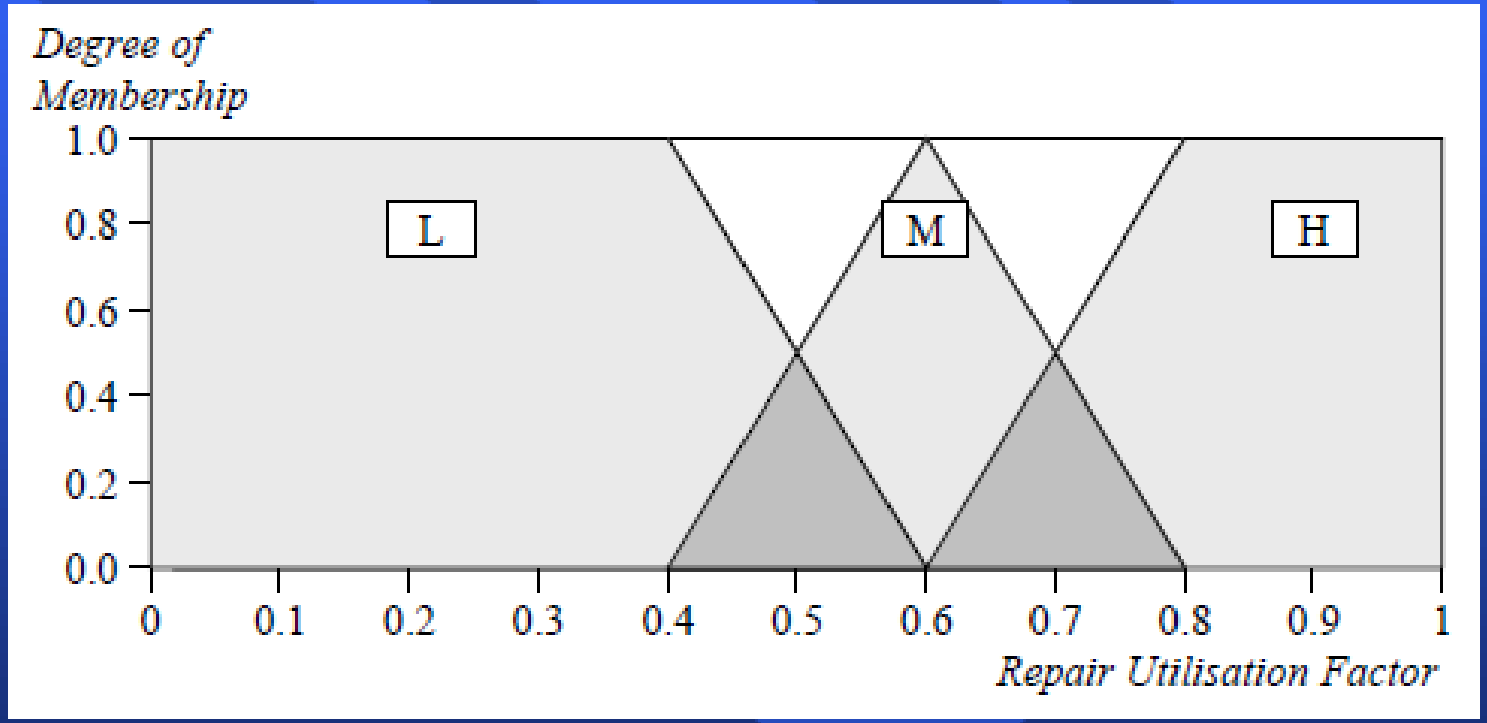
Fuzzy sets of *Mean Delay m*



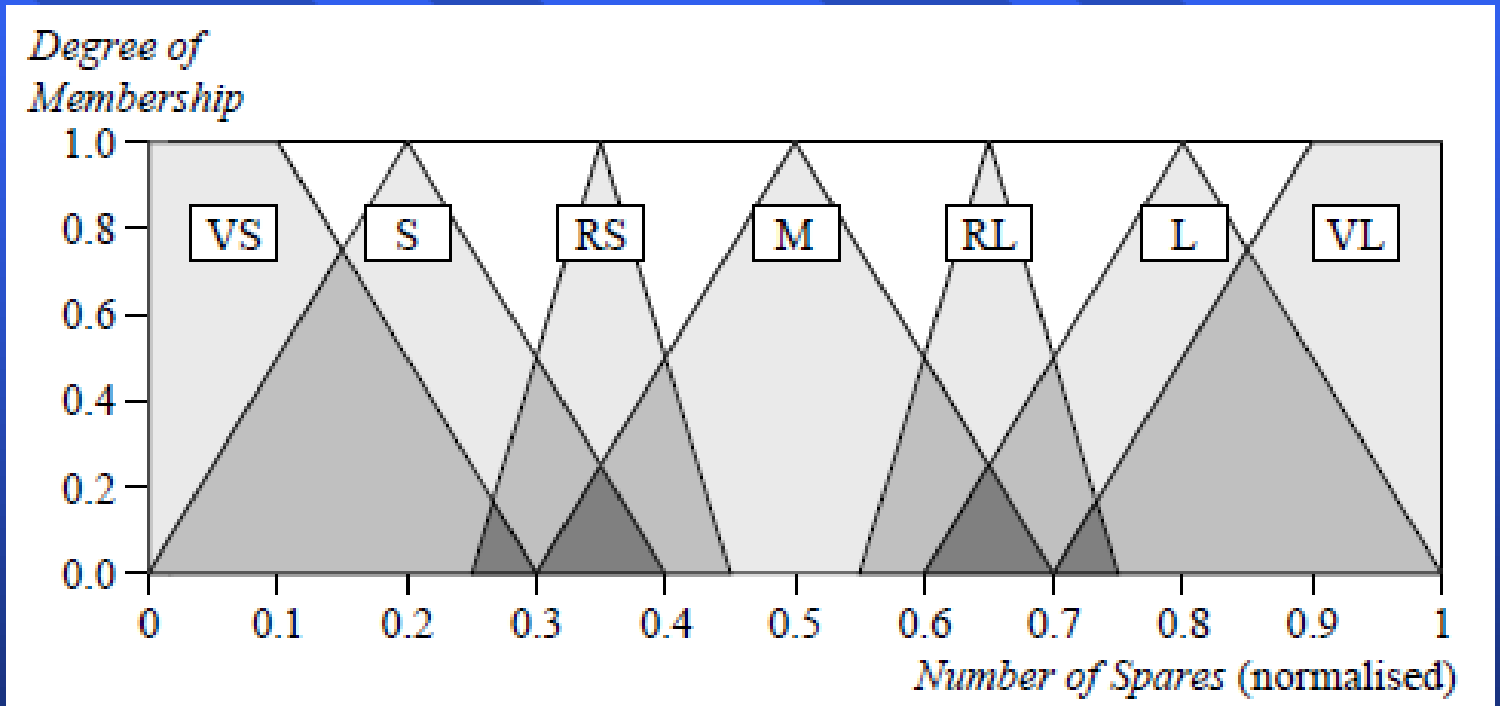
Fuzzy sets of *Number of Servers s*



Fuzzy sets of *Repair Utilisation Factor* ρ



Fuzzy sets of *Number of Spares n*



Step 3: Elicit and construct fuzzy rules

To accomplish this task, we might ask the expert to describe how the problem can be solved using the fuzzy linguistic variables defined previously.

Required knowledge also can be collected from other sources such as books, computer databases, flow diagrams and observed human behaviour.

The rule table

Rule	*m*	*s*	ρ	*n*	Rule	*m*	*s*	ρ	*n*	Rule	*m*	*s*	ρ	*n*
1	VS	S	L	VS	10	VS	S	M	S	19	VS	S	H	VL
2	S	S	L	VS	11	S	S	M	VS	20	S	S	H	L
3	M	S	L	VS	12	M	S	M	VS	21	M	S	H	M
4	VS	M	L	VS	13	VS	M	M	RS	22	VS	M	H	M
5	S	M	L	VS	14	S	M	M	S	23	S	M	H	M
6	M	M	L	VS	15	M	M	M	VS	24	M	M	H	S
7	VS	L	L	S	16	VS	L	M	M	25	VS	L	H	RL
8	S	L	L	S	17	S	L	M	RS	26	S	L	H	M
9	M	L	L	VS	18	M	L	M	S	27	M	L	H	RS

Rule Base 1

1. If (utilisation_factor is L) then (number_of_spares is S)
2. If (utilisation_factor is M) then (number_of_spares is M)
3. If (utilisation_factor is H) then (number_of_spares is L)

4. If (mean_delay is VS) and (number_of_servers is S) then (number_of_spares is VL)
5. If (mean_delay is S) and (number_of_servers is S) then (number_of_spares is L)
6. If (mean_delay is M) and (number_of_servers is S) then (number_of_spares is M)

7. If (mean_delay is VS) and (number_of_servers is M) then (number_of_spares is RL)
8. If (mean_delay is S) and (number_of_servers is M) then (number_of_spares is RS)
9. If (mean_delay is M) and (number_of_servers is M) then (number_of_spares is S)

10. If (mean_delay is VS) and (number_of_servers is L) then (number_of_spares is M)
11. If (mean_delay is S) and (number_of_servers is L) then (number_of_spares is S)
12. If (mean_delay is M) and (number_of_servers is L) then (number_of_spares is VS)

Step 4: Encode the fuzzy sets, fuzzy rules and procedures to perform fuzzy inference into the expert system

To accomplish this task, we may choose one of two options: to build our system using a programming language such as C/C++ or Pascal, or to apply a fuzzy logic development tool such as MATLAB Fuzzy Logic Toolbox or Fuzzy Knowledge Builder.

Step 5: Evaluate and tune the system

The last, and the most laborious, task is to evaluate and tune the system. We want to see whether our fuzzy system meets the requirements specified at the beginning.

Several test situations depend on the mean delay, number of servers and repair utilisation factor.

The Fuzzy Logic Toolbox can generate surface to help us analyse the system's performance.

Tuning fuzzy systems

1. Review model input and output variables, and if required redefine their ranges.
2. Review the fuzzy sets, and if required define additional sets on the universe of discourse. The use of wide fuzzy sets may cause the fuzzy system to perform roughly.
3. Provide sufficient overlap between neighbouring sets. It is suggested that triangle-to-triangle and trapezoid-to-triangle fuzzy sets should overlap between 25% to 50% of their bases.

4. Review the existing rules, and if required add new rules to the rule base.
5. Examine the rule base for opportunities to write hedge rules to capture the pathological behaviour of the system.
6. Adjust the rule execution weights. Most fuzzy logic tools allow control of the importance of rules by changing a weight multiplier.
7. Revise shapes of the fuzzy sets. In most cases, fuzzy systems are highly tolerant of a shape approximation.

Review questions

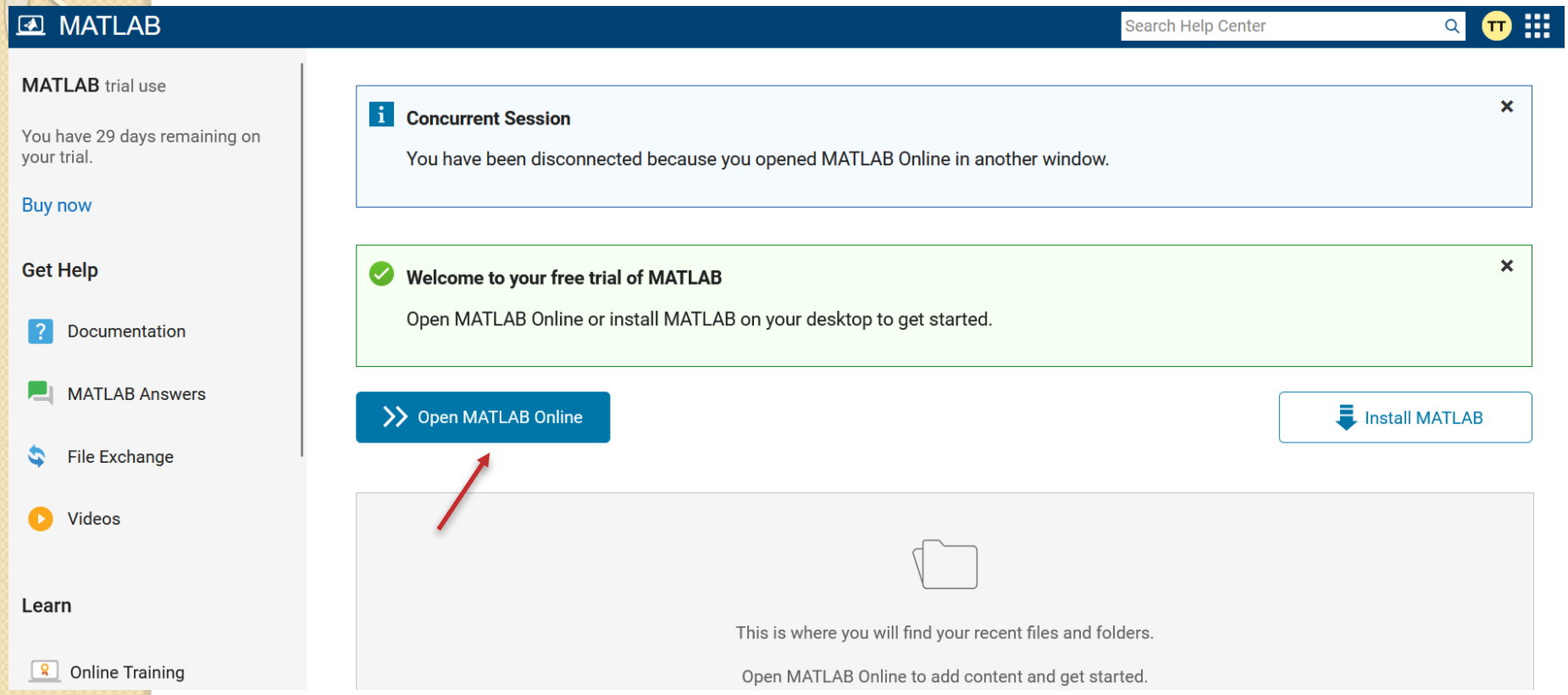
- What is expert system?
- What is uncertainty?
- What is fuzzy expert system?
- What type of problem is suitable to be solved using fuzzy expert system?

Demo using Matlab (Lab Session)

- How to construct Fuzzy Expert System using Fuzzy Logic toolbox
  - Part 1: https://www.mathworks.com/videos/getting-started-with-fuzzy-logic-toolbox-part-1-68764.html
  - Part 2: https://www.mathworks.com/videos/getting-started-with-fuzzy-logic-toolbox-part-2-68765.html
  - Part 3: https://www.mathworks.com/videos/getting-started-with-fuzzy-logic-toolbox-part-3-68766.html

- Go to matlab online: https://matlab.mathworks.com/
- In command prompt: fuzzyLogicDesigner
- It will open the fuzzy logic toolbox

Open MATLAB Online



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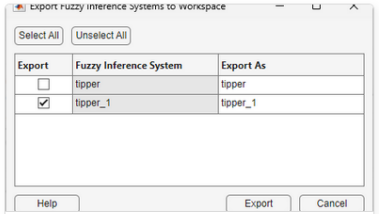
Type

- All 4
- MATLAB 4
- Simulink 0

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Get Started with Fuzzy Logic Toolbox — Examples

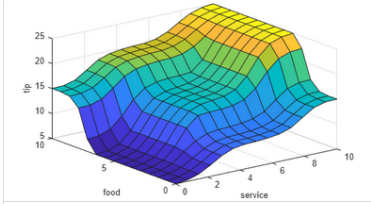
R2025b



Build Fuzzy Systems Using Fuzzy Logic Designer

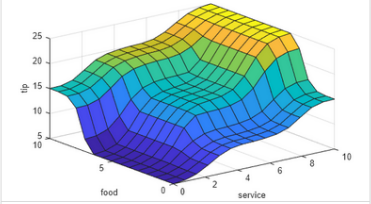
Interactively construct a fuzzy inference system using the Fuzzy Logic Designer app.

Since R2022b



Build Fuzzy Systems at the Command Line

Construct a fuzzy inference system at the MATLAB command line.



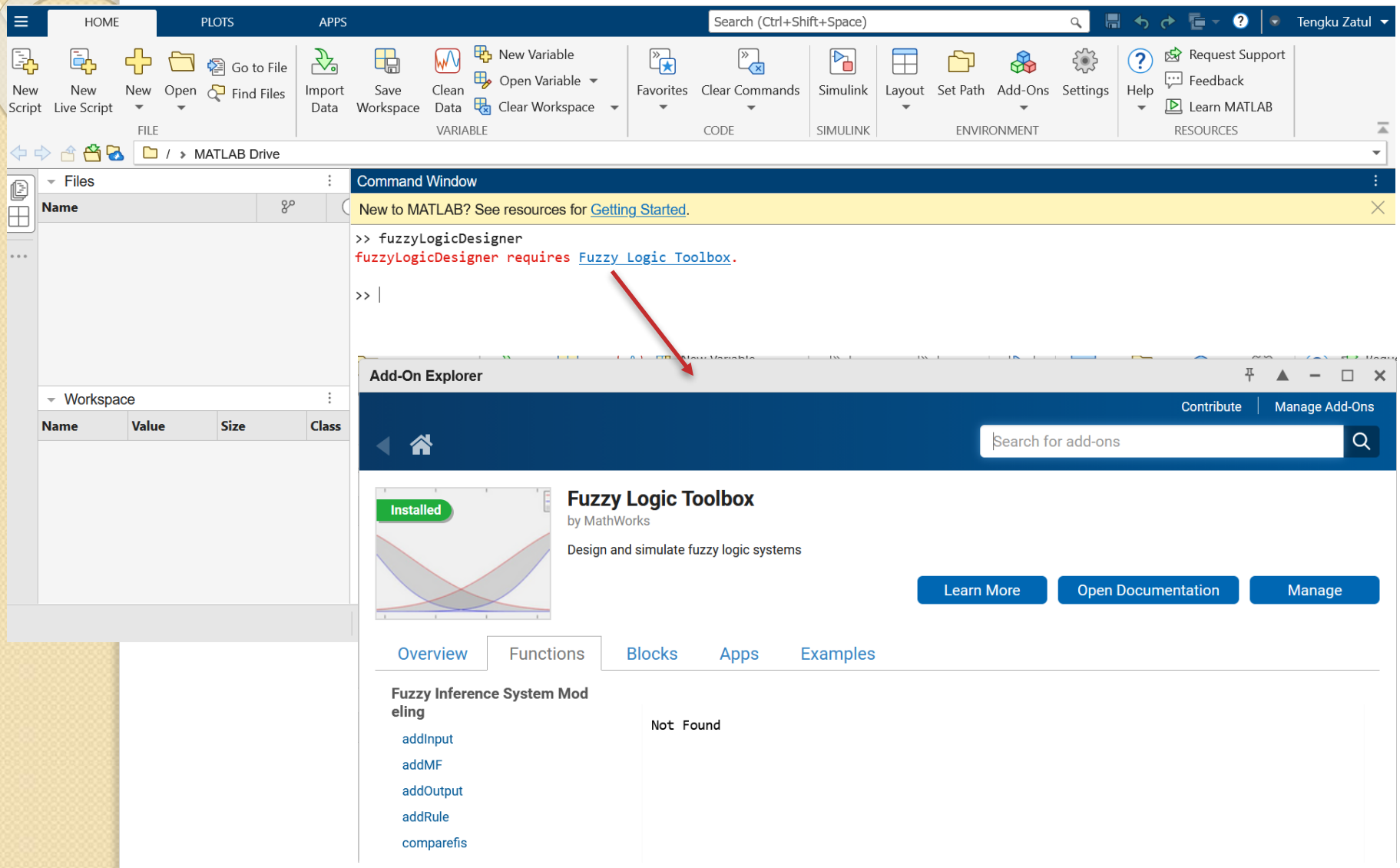
Fuzzy vs. Nonfuzzy Logic

To illustrate the value of fuzzy logic, examine both linear and fuzzy approaches to a basic tipping problem.

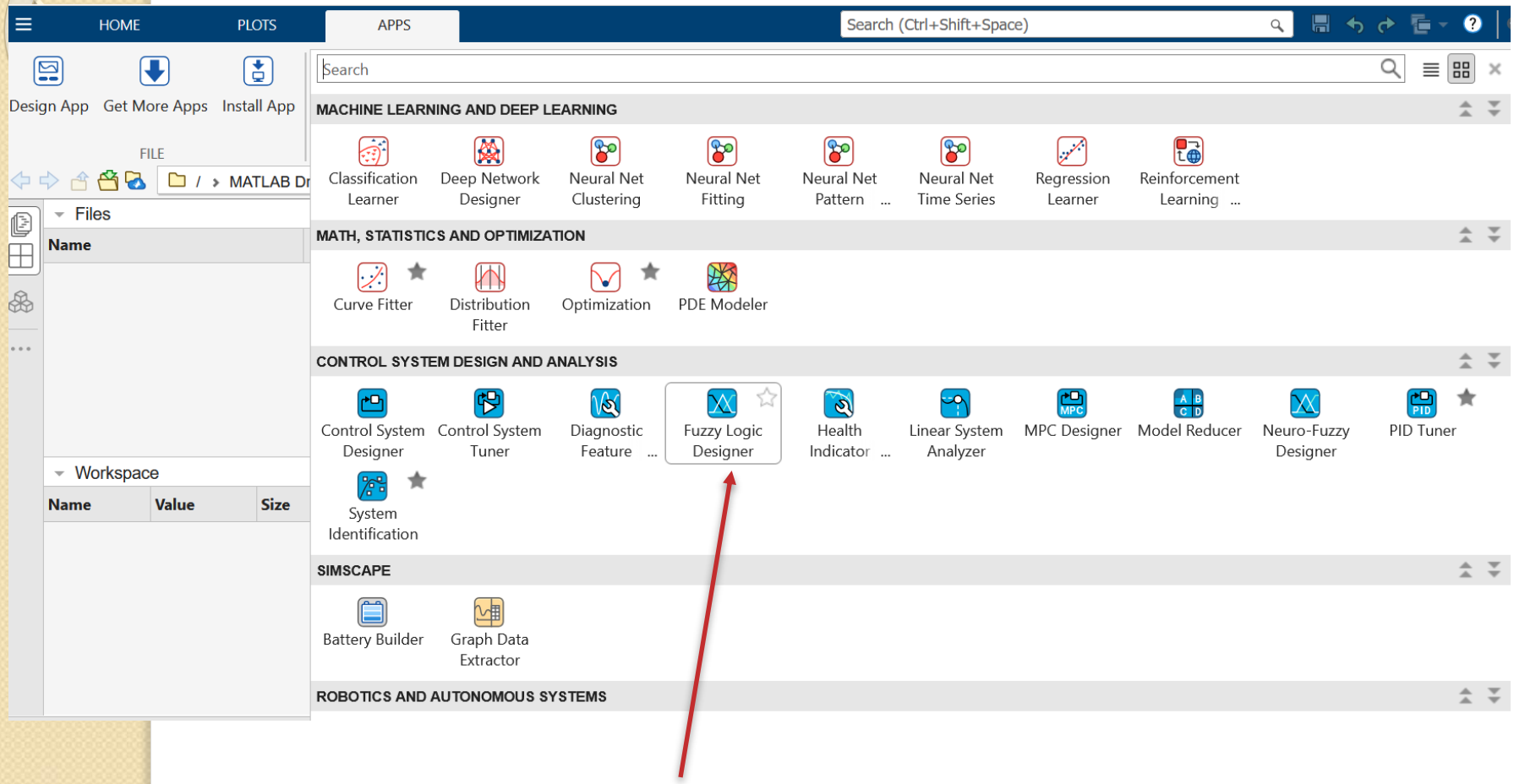


www.mathworks.com/help/fuzzy/examples.html?category=getting-started-with-fuzzy-logic-toolbox&exampleproduct=all&newonly=&s_tid=CRUX_lftnav

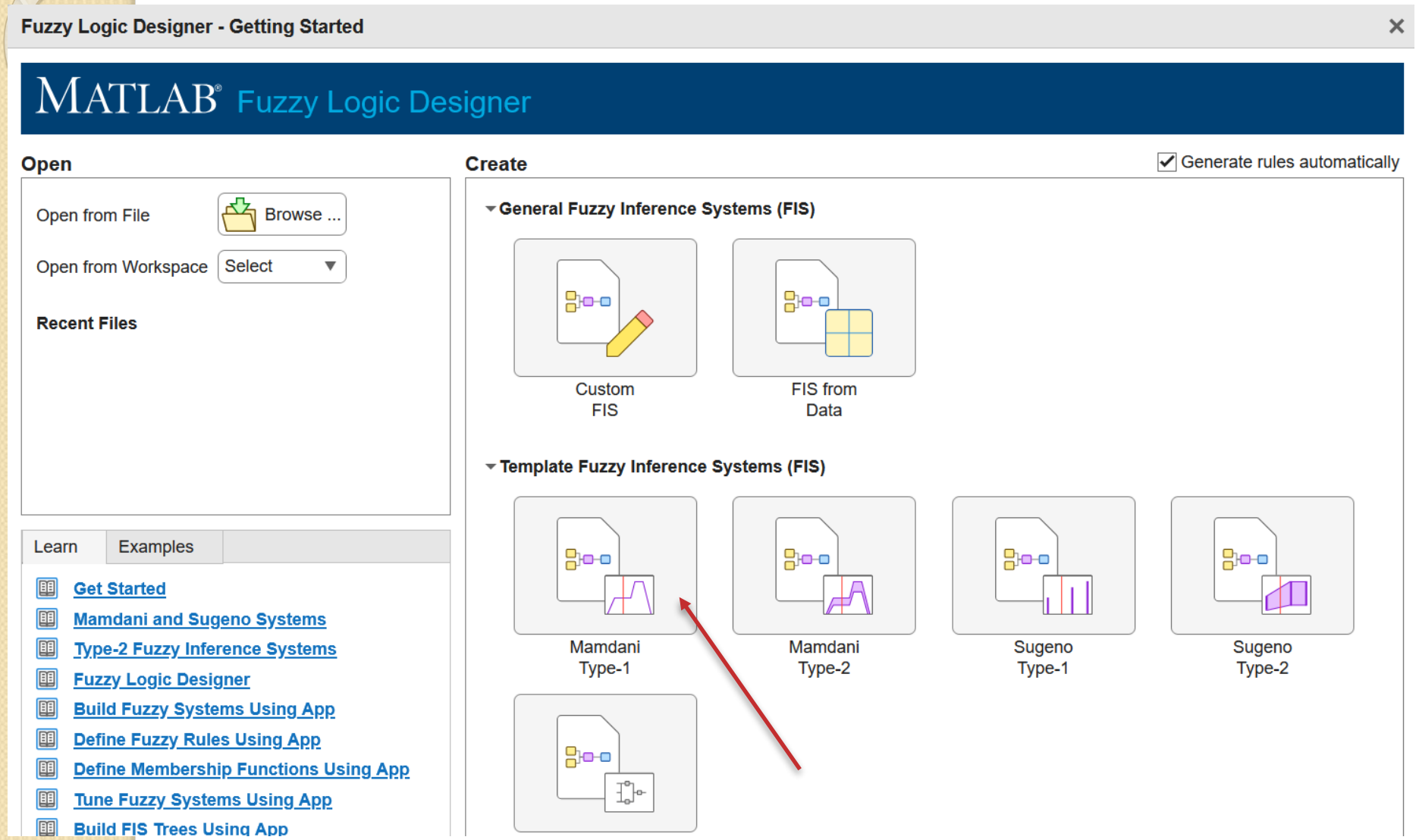
fuzzyLogicDesigner



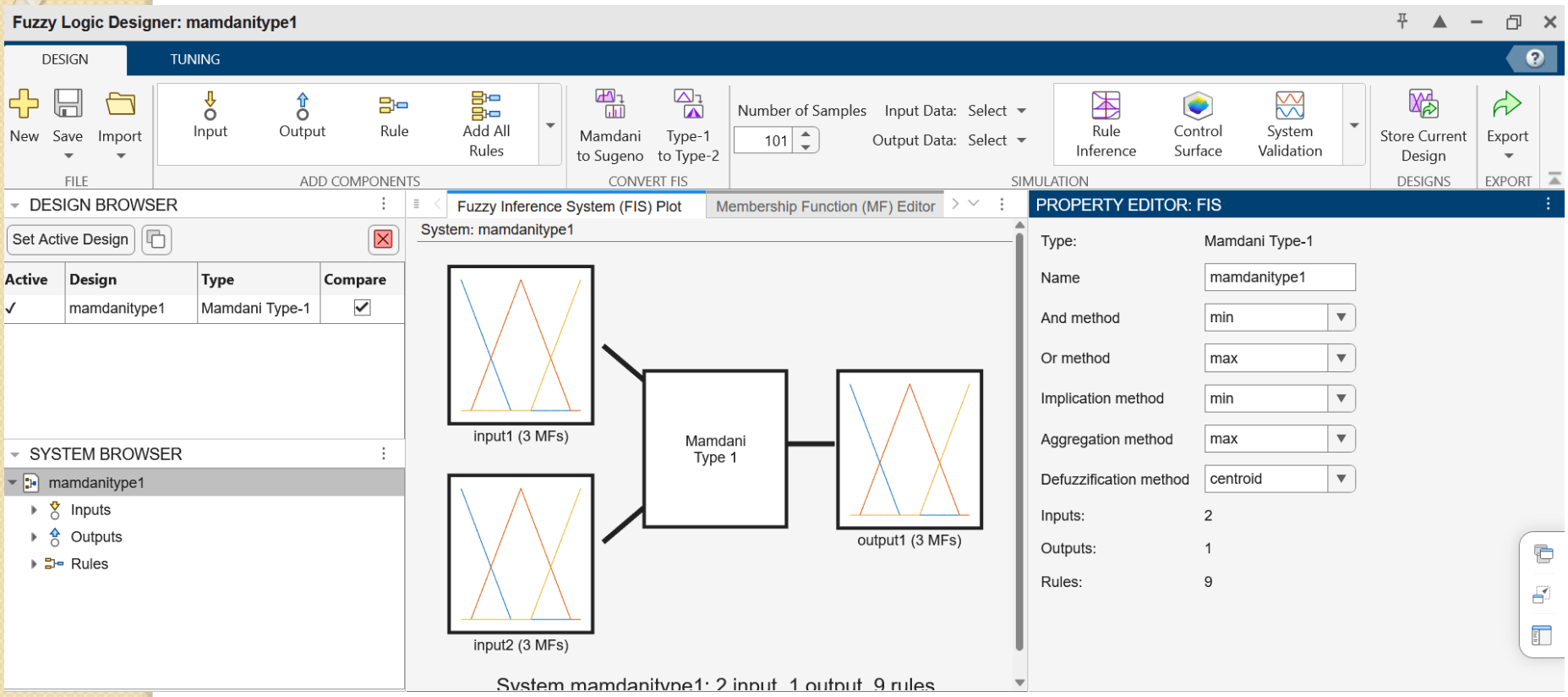
Fuzzy Logic Designer



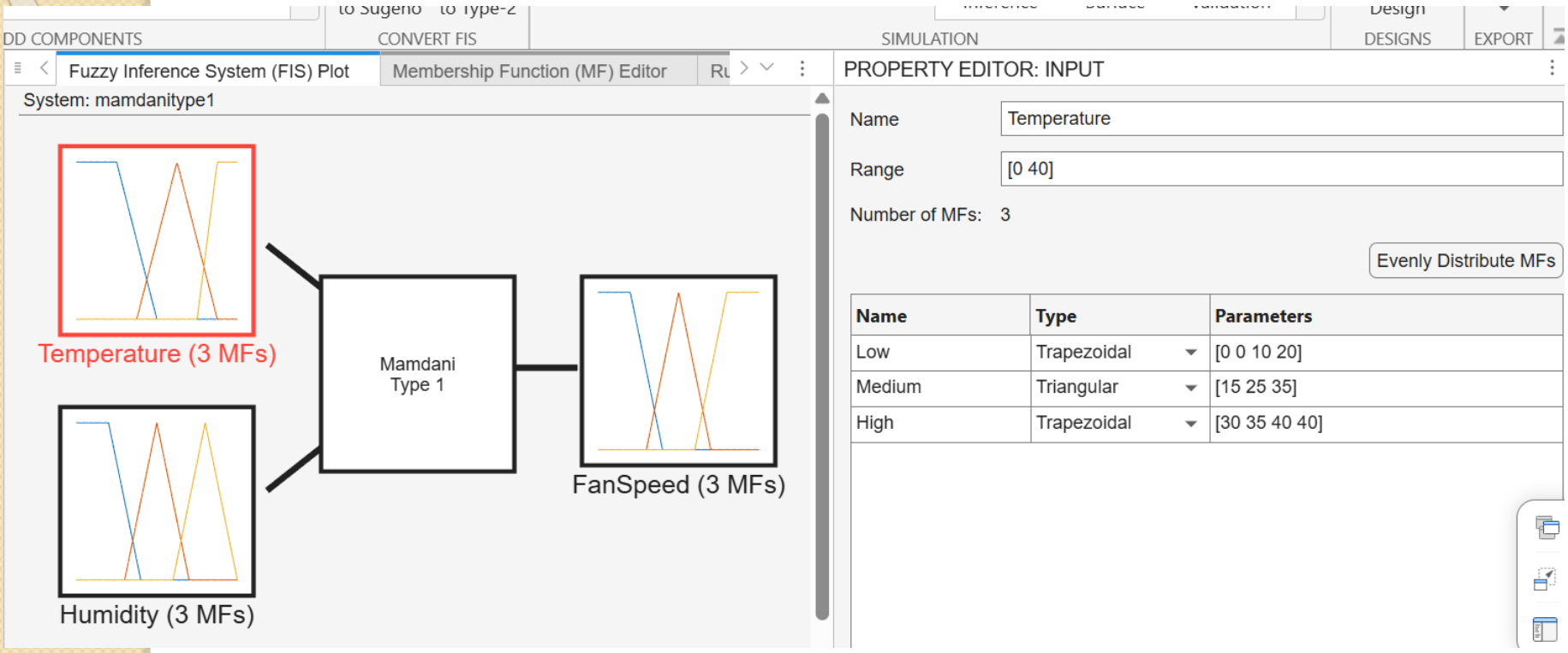
Create a simple FIS



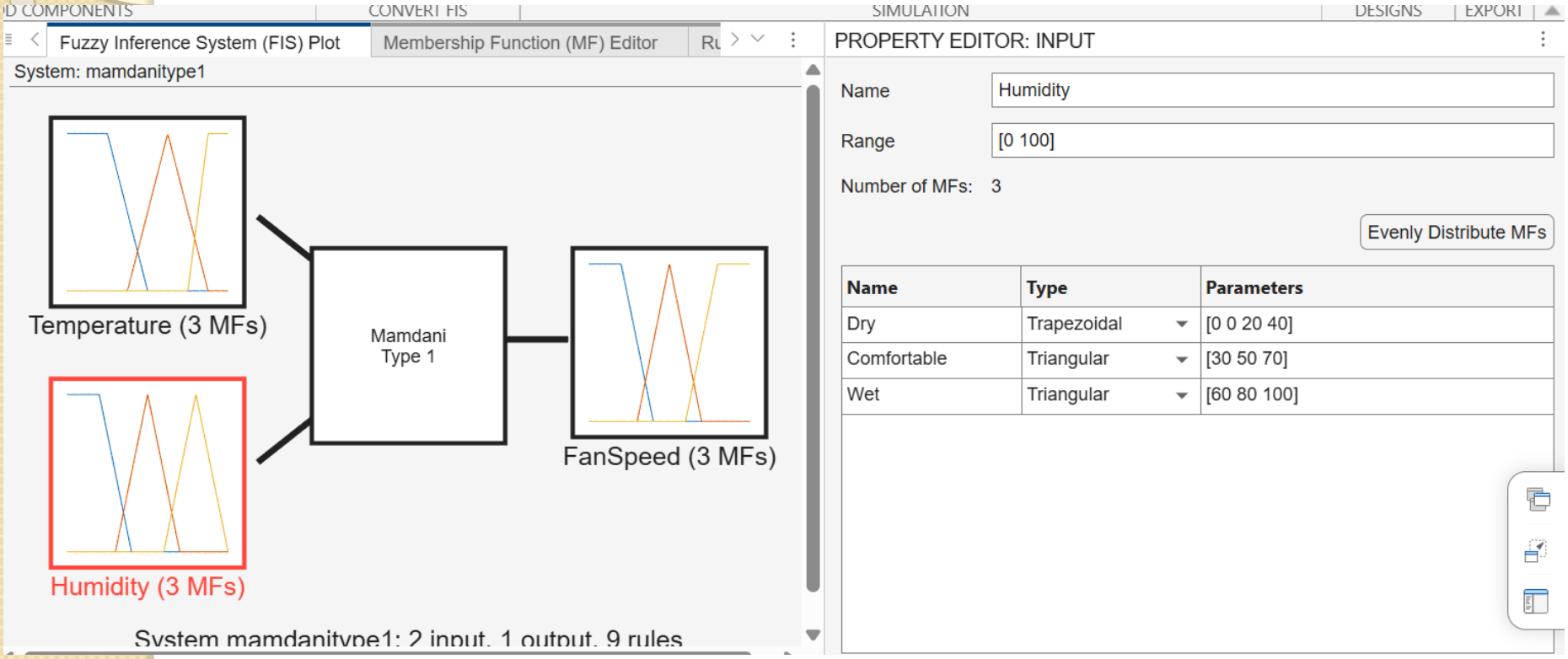
Mamdani Type 1



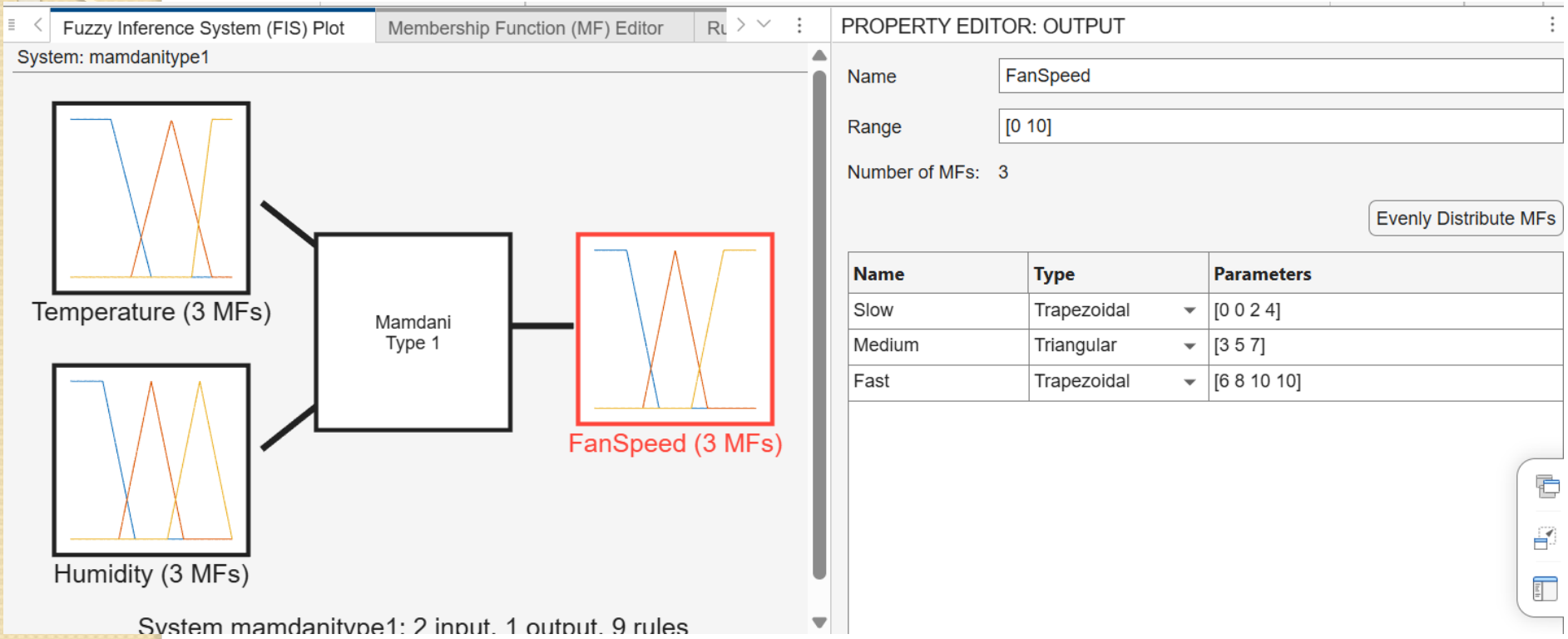
Define Input Temperature



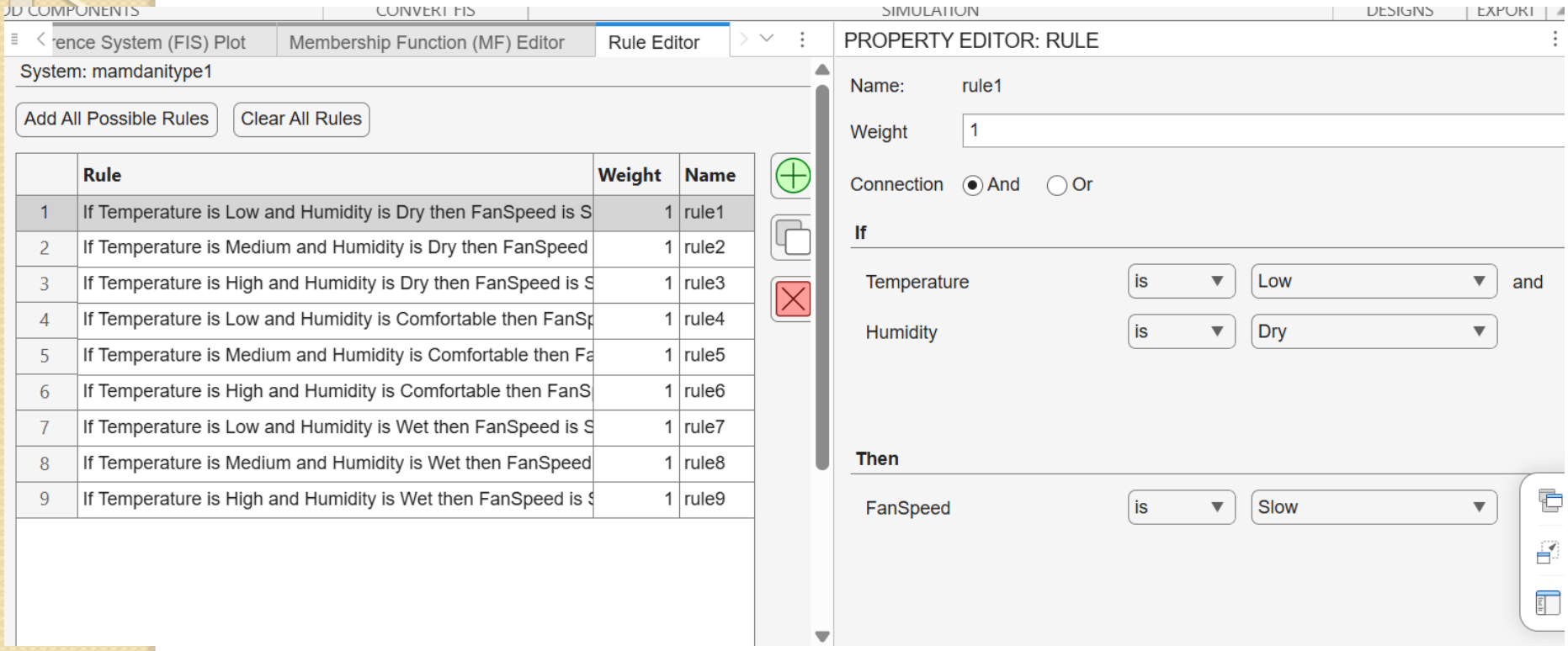
Define Input Humidity



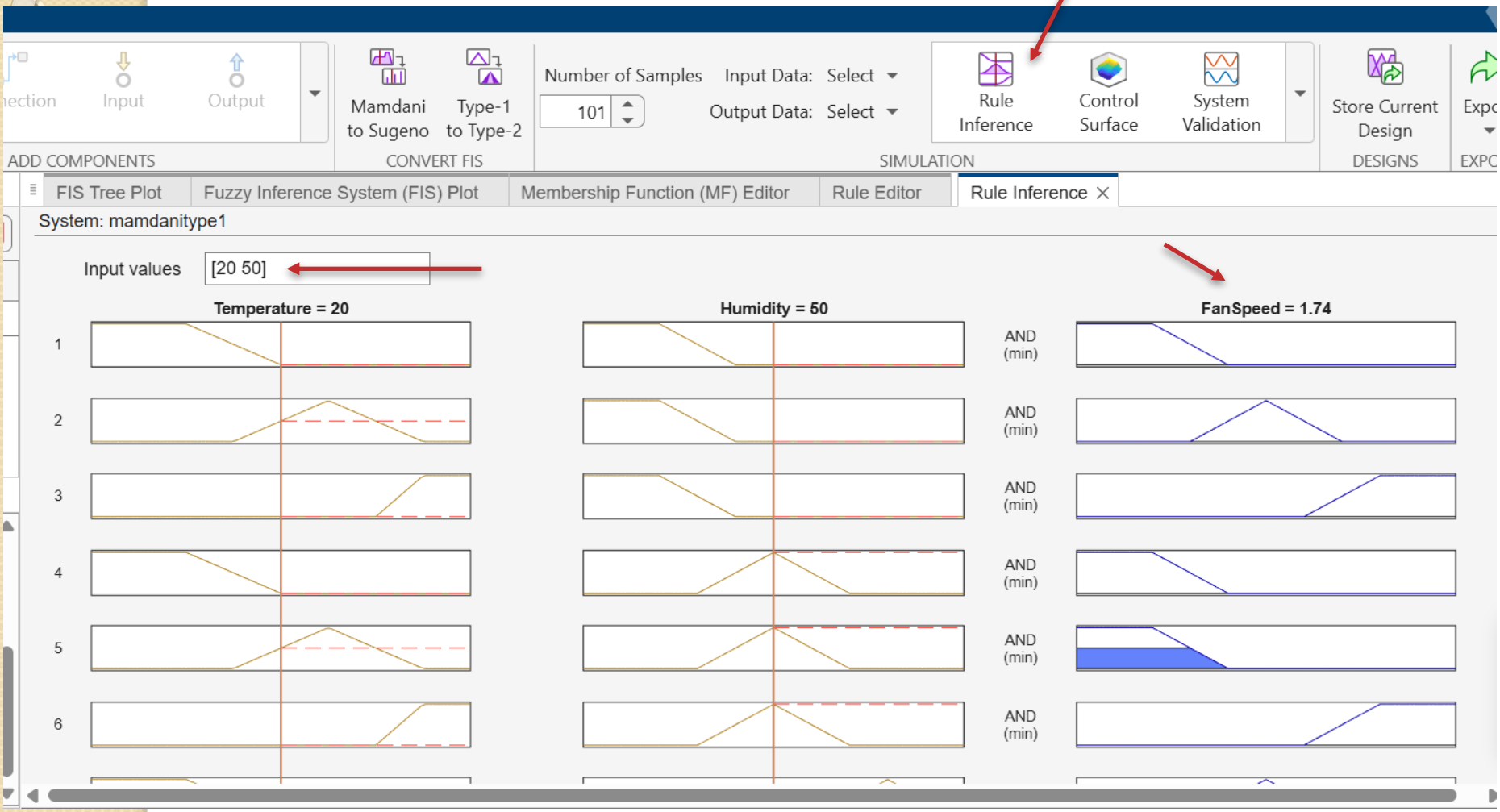
Define Output Fan Speed



Rule Editor


ence System (FIS) Plot
Membership Function (MF) Editor
Rule Editor
System: mamdanitype1
Add All Possible Rules
Clear All Rules
Rule
Weight
Name
1 If Temperature is Low and Humidity is Dry then FanSpeed is S 1 rule1
2 If Temperature is Medium and Humidity is Dry then FanSpeed 1 rule2
3 If Temperature is High and Humidity is Dry then FanSpeed is S 1 rule3
4 If Temperature is Low and Humidity is Comfortable then FanSp 1 rule4
5 If Temperature is Medium and Humidity is Comfortable then Fa 1 rule5
6 If Temperature is High and Humidity is Comfortable then FanS 1 rule6
7 If Temperature is Low and Humidity is Wet then FanSpeed is S 1 rule7
8 If Temperature is Medium and Humidity is Wet then FanSpeed 1 rule8
9 If Temperature is High and Humidity is Wet then FanSpeed is S 1 rule9
PROPERTY EDITOR: RULE
Name: rule1
Weight 1
Connection And Or
If
Temperature is Low and
Humidity is Dry
Then
FanSpeed is Slow

Rule Inference



Control Surface

